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IC103

Introductory Economics for Business and Finance

Lecture 2: Elasticity and Consumer Choice

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Reading

Relevant chapters and sections for Lecture 2

Core textbook

Sloman, J., Wride, A. and Garratt, D. (2018). Economics. 10th edition.

Section 2.4

Background to demand and consumer choice

Section 4.1

Except Box 4.3 and threshold concept 8. Boxes 4.1 and 4.2 are optional.

Section 4.2

Except “Looking at the Maths” on page 114.

Section 4.3

Except Box 4.4 and Box 4.5. “Looking at the Maths” pp. 119 is optional.

You do not need to read every optional box. Focus on the lecture material and seminar questions.

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Section 1: How Markets Work — Demand and Supply

Price Elasticity of Demand

Elasticity

How much buyers and sellers respond to changes in market conditions

Elasticity

Elasticity measures **how much** buyers and sellers respond to changes in market conditions.

Examples: price and income.

Price elasticity of demand

The responsiveness of **quantity demanded to a change in price.**

Basic idea

Price elasticity of demand is computed as the percentage change in quantity demanded divided by the percentage change in price.

Elasticity tells us how sensitive demand is to a change in price.

Price Elasticity of Demand

Formula and interpretation

Formula

$$P_{\varepsilon_D} = \frac{\%\Delta Q_D}{\%\Delta P}$$

Where ε_D is the elasticity of the demand, $P\varepsilon_D$ the price elasticity of demand, $\%\Delta Q_D$ is the % change in the quantity demanded, $\%\Delta P$ is the % change in the price

Sign

Price elasticity of demand is normally negative because demand curves slope downwards.

Use the absolute value

PεD > 1: elastic demand
PεD = 1: unit elastic demand
PεD < 1: inelastic demand

In practice, we usually discuss the absolute value of elasticity.

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If the price of a product rises by 10% and quantity demanded falls by 20%, demand is:



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What affects price elasticity of demand?

Key determinants

Availability of close substitutes

Goods or services with close substitutes are usually more elastic. When price rises, consumers can switch to the substitute.

Necessities vs. luxuries

Necessities are usually more inelastic.

Luxuries are usually more elastic.

Example

If the price of food rises, quantity demanded may not change very much.

If the price of a luxury item rises, quantity demanded may fall by much more.

The easier it is to avoid or replace a good, the more elastic demand tends to be.

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Which product is likely to have the most price-elastic demand?

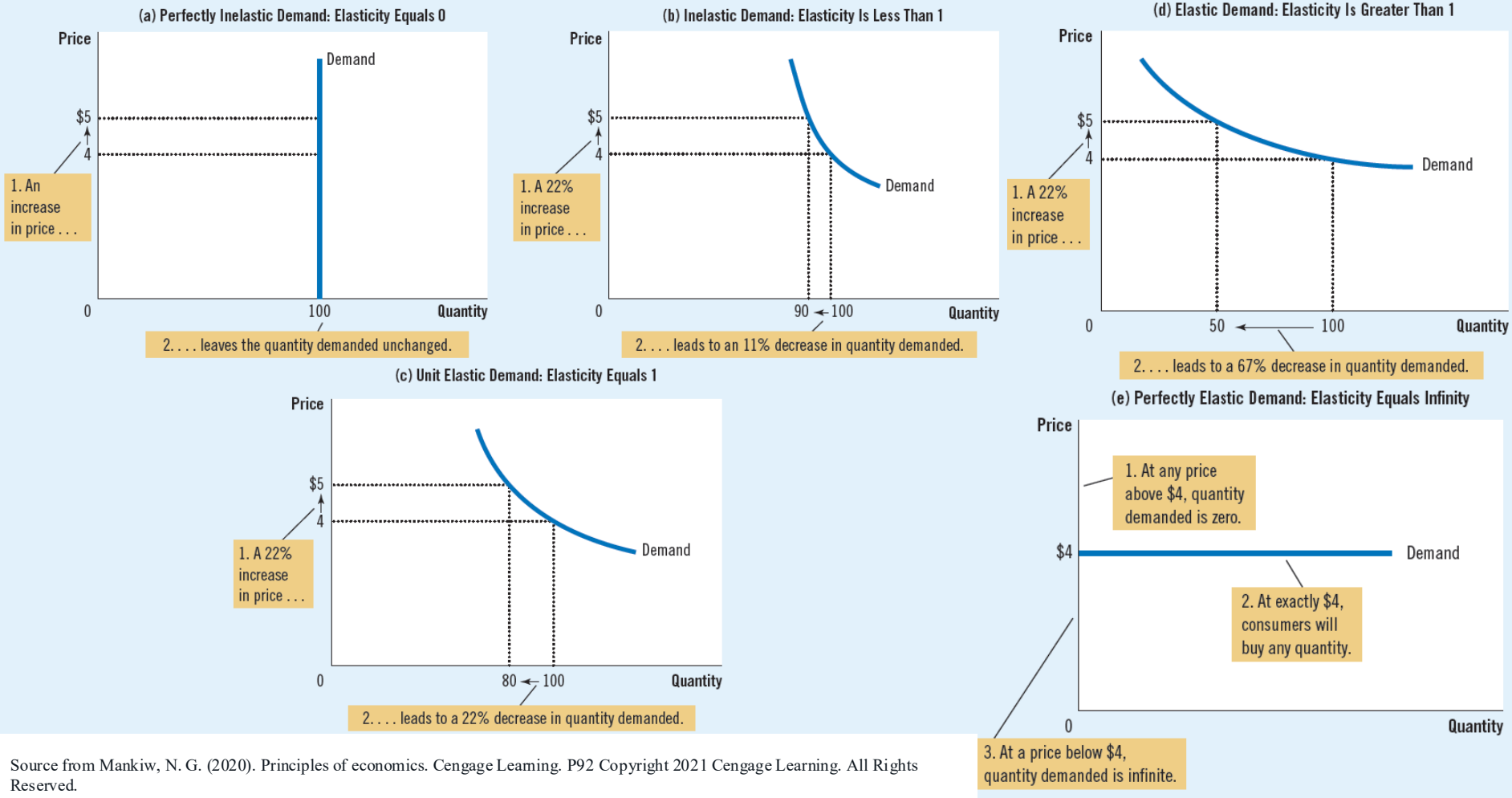
0	0	0	0
Prescription medicine	Electricity	A particular brand of coffee	Tap water

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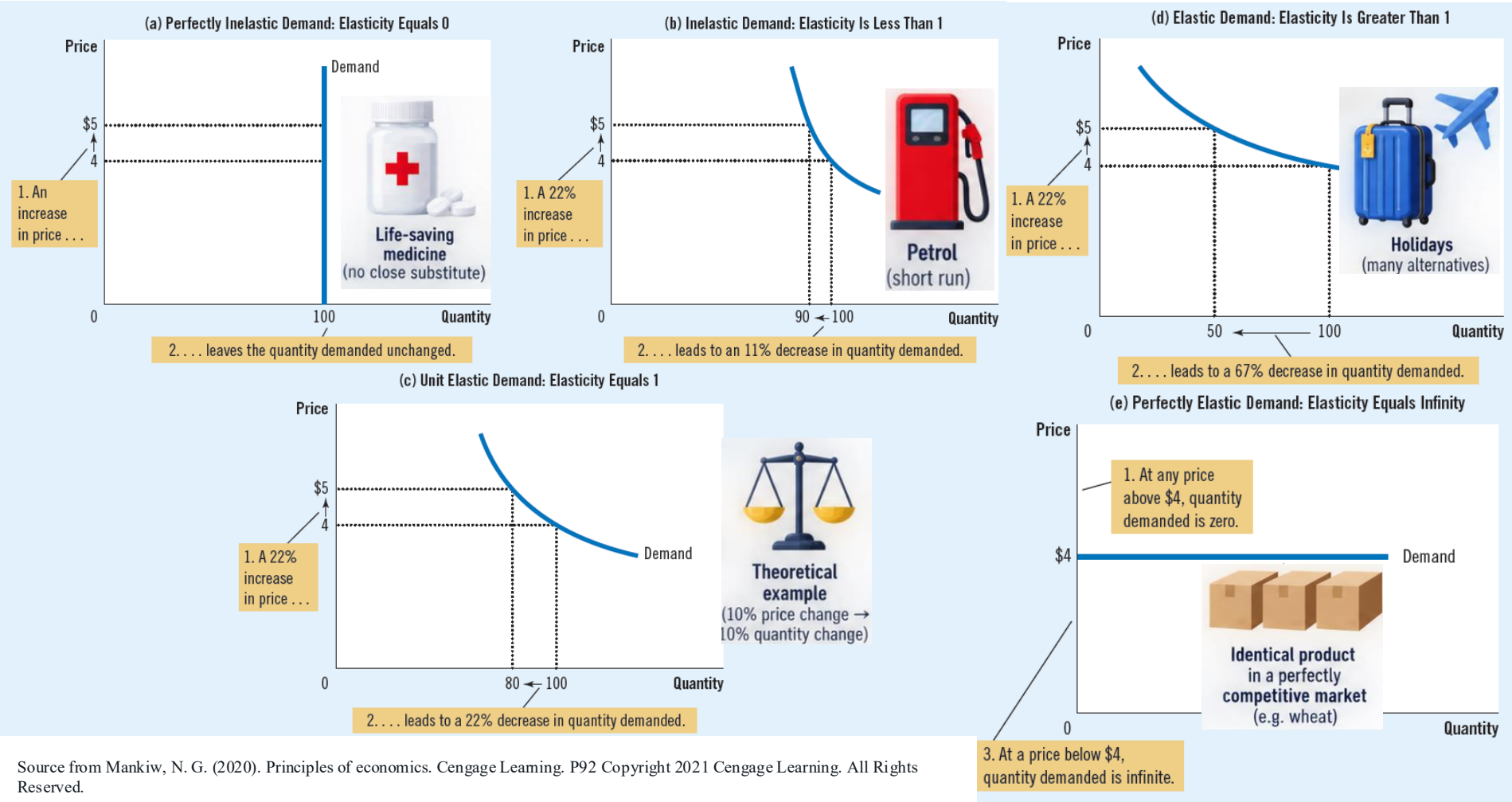
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Types of Demand Elasticity



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Types of Demand Elasticity: Examples



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Measuring Elasticity

- Measure elasticity at two points
  - **Percentage method**
    Move from point 1 to 2

$$P_{\varepsilon_D} = \frac{(Q_2 - Q_1)/Q_1}{(P_2 - P_1)/P_1}$$

  - **Arc elasticity-midpoint method**
    Elasticity between point 1 and point 2

$$P_{\varepsilon_D} = \frac{(Q_2 - Q_1)/[\frac{Q_2 + Q_1}{2}]}{(P_2 - P_1)/[\frac{P_2 + P_1}{2}]}$$

- Measure elasticity at a single point
  - **Point elasticity**

$$P_{\varepsilon_D} = \frac{dQ_D}{dP} * \frac{P}{Q}$$

Measuring Elasticity at Two Points

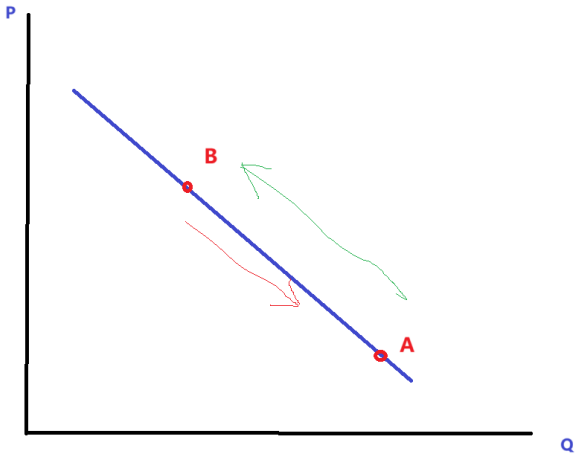
Arc elasticity — midpoint method to calculate percentage change

- We cannot normally talk about the elasticity of the whole curve. Elasticity can differ across the curve
- We can talk about the elasticity between two points though
- *Issue: Point A: Price= £4 Quantity= 120; Point B: Price= £6 Quantity= 80*

Elasticity from point A to point B = $\frac{(80-120)/120}{(6-4)/4}$ *= -0.66 ; Elasticity from point B to point A =* $\frac{(120-80)/80}{(4-6)/6}$ *= -1.5*

- **Solution : midpoint method – to calculate percentage change**

$$P_{\varepsilon_D} = \frac{(Q_2 - Q_1)/[\frac{Q_2 + Q_1}{2}]}{(P_2 - P_1)/[\frac{P_2 + P_1}{2}]}$$



Example: Elasticity from point A to point B = $\frac{(80-120)/100}{(6-4)/5}$ *= -1*

Measuring Elasticity at a Single Point

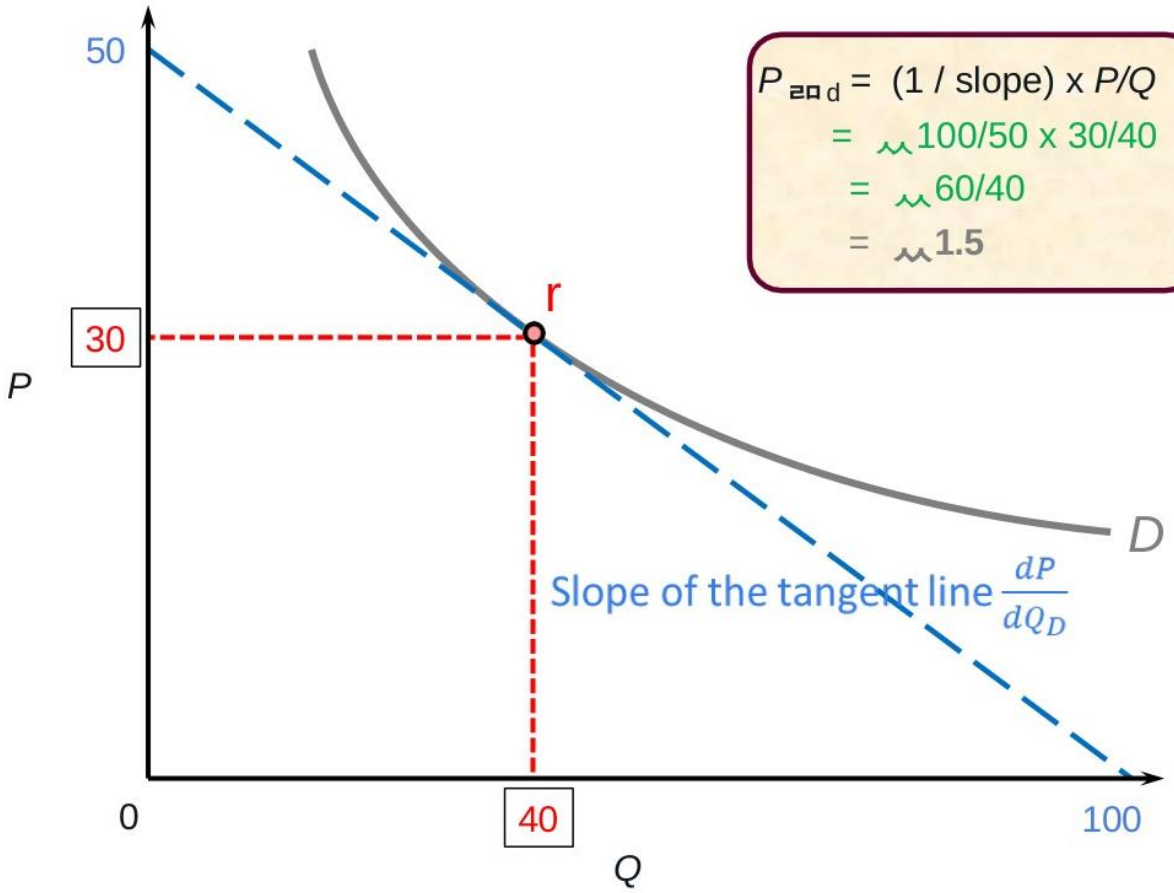
Point elasticity

- Recall $P_{\varepsilon_D} = \frac{\%\Delta Q_D}{\%\Delta P} = \frac{\Delta Q_D / Q_D}{\Delta P / P} = \frac{\Delta Q_D}{\Delta P} * \frac{P}{Q}$
- At a single point, change is infinitesimally small

$$P_{\varepsilon_D} = \frac{dQ_D}{dP} * \frac{P}{Q}$$

- $\frac{dQ_D}{dP}$ = how sensitive quantity demanded is to a very small change in price around that point
  - the first derivative of demand function $Q_D = f(P)$
  - the demand curve (Inverse demand function $P = f(Q_D)$, Y axis: price; X axis: quantity); slope $\frac{dP}{dQ_D}$ is found by drawing a tangent to curve at the point
  - Demand curve $P_{\varepsilon_D} = \frac{1}{slope} * \frac{P}{Q}$

Point Elasticity: Graphical Example

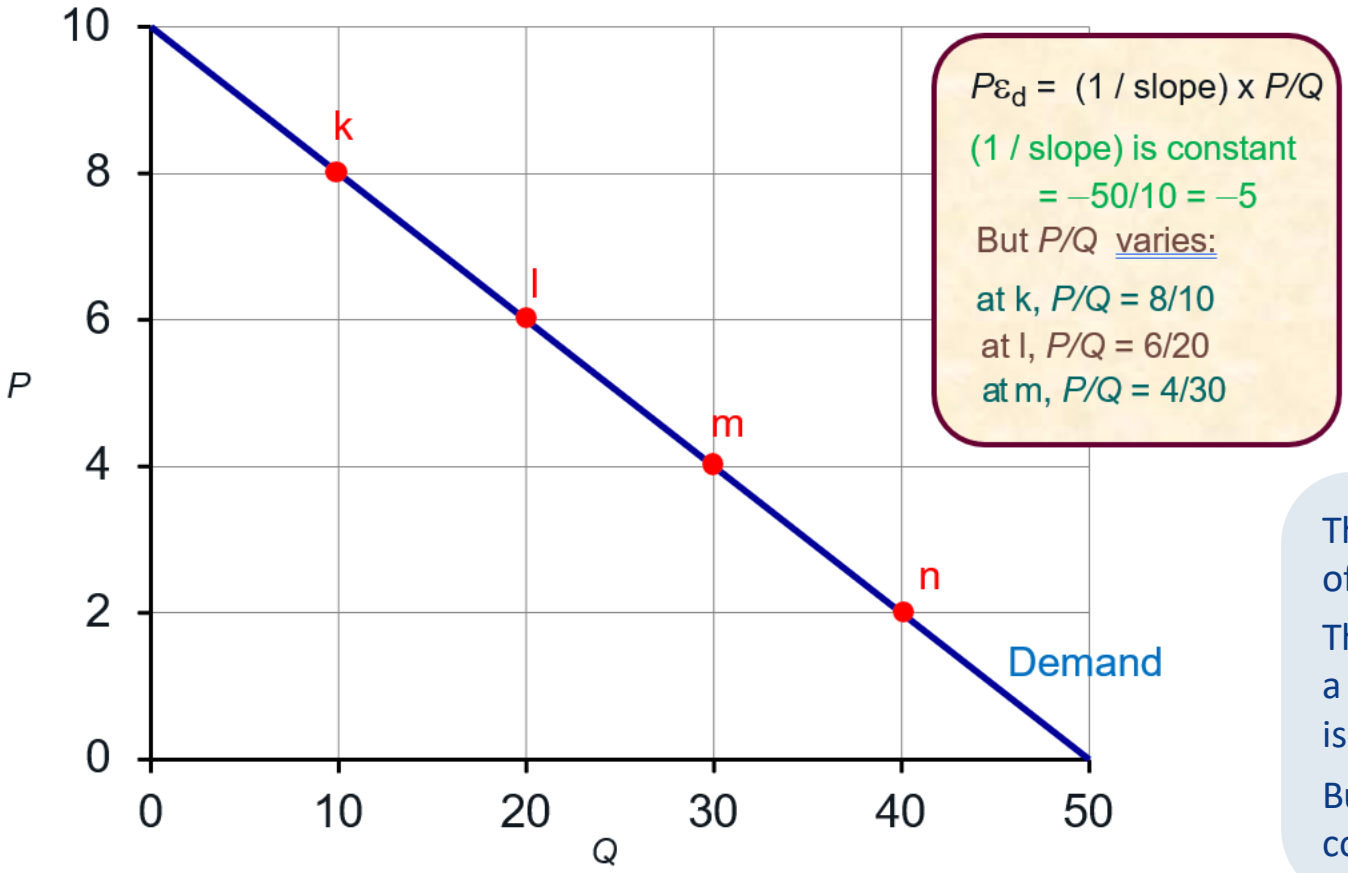


P_{ε_D} = (1 / slope) x P/Q
= 100/50 x 30/40
= 60/40
= 1.5

Because $P_{\varepsilon_D} > 1$, demand at this point is elastic.

And this brings us to an important result: elasticity can actually change as we move along the same demand curve.

Different Elasticities Along a Straight-Line Demand Curve



$P\varepsilon_d$ = (1 / slope) x P/Q

(1 / slope) is constant
= –50/10 = –5

But P/Q varies:

at k, P/Q = 8/10
at l, P/Q = 6/20
at m, P/Q = 4/30

This is something students often find surprising.

The demand curve here is a straight line, so its slope is constant.

But its elasticity is not constant.

Why Elasticity Matters

Practical question: why should a business or policymaker care about elasticity?

- **Pricing strategy**

  For a firm, elasticity matters when setting prices. A price cut only makes sense if the increase in sales is large enough to compensate for receiving less money per unit.

- **Public policy**

  If demand for a product is very inelastic, increasing its price through taxation may not reduce consumption very much.
  E.g. Cigarettes. Price ↑ 10% → Quantity demanded ↓ only 2%

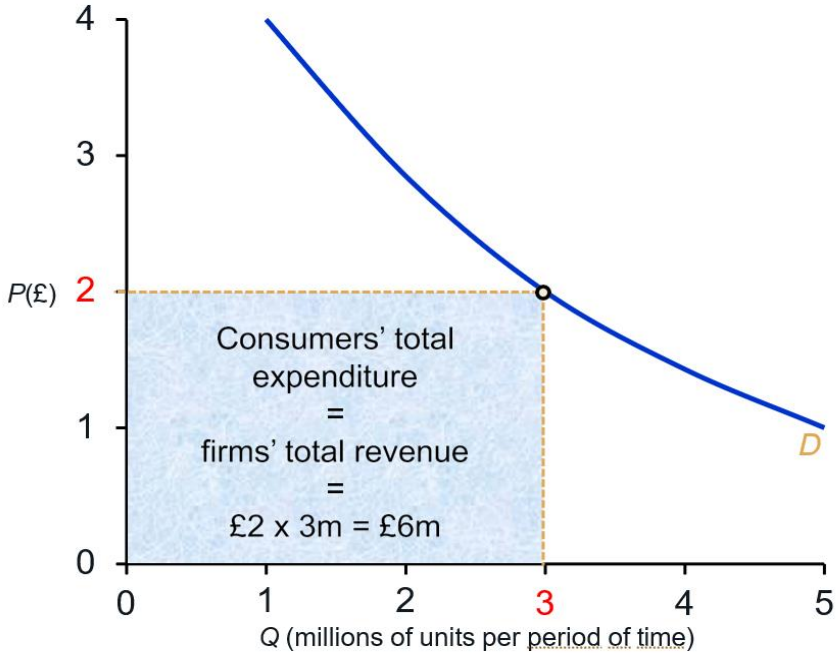
Total consumer expenditure *(TE) or* Total revenue received by firms *(TR)*

- Computed as a multiplication of price and quantity purchased (sold by firms)

$$TE = TR = P * Q$$

Total Consumer Expenditure / Total Revenue

- Consumers' **total expenditure (TE)** is the amount spent on a good: $TE = P \times Q$
- It is the area of the rectangle formed by price and quantity purchased.



Consumer expenditure = Firm revenue

In this simple setting:

$$\boldsymbol{TE = TR = P \times Q}$$

Example

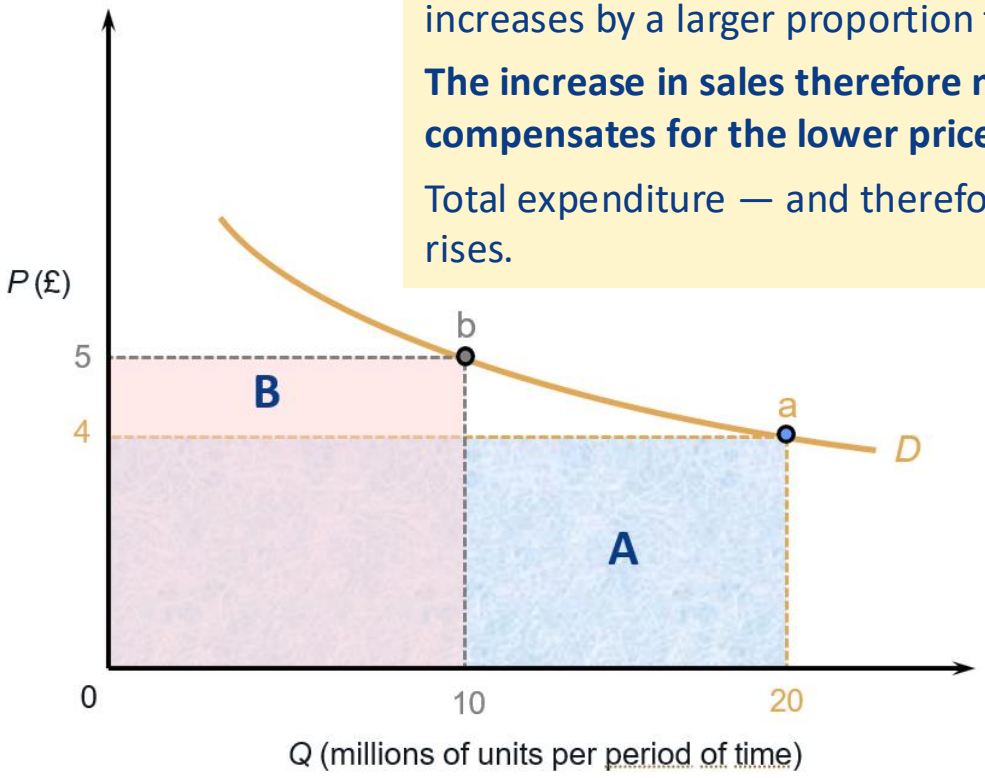
If $P = £2$ and $Q = 3$ million, total consumer expenditure:

$$TE = TR = £6\,million$$

This £6 million is also the total revenue received by firms.

Elastic Demand and Expenditure

Elastic demand (P_{ϵ_D}>1)



Suppose price falls from £5 to £4. That means the firm receives less money for each unit sold.

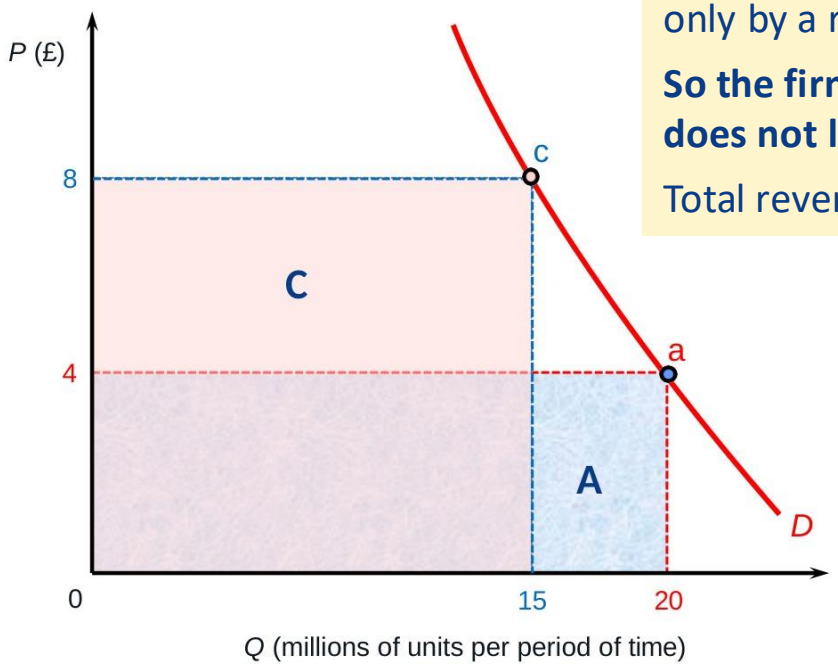
But because demand is elastic, quantity demanded increases by a larger proportion than the fall in price.

The increase in sales therefore more than compensates for the lower price.

Total expenditure — and therefore total revenue — rises.

Inelastic Demand and Expenditure

Inelastic demand (P_{ϵ_D}<1)



Suppose price rises.

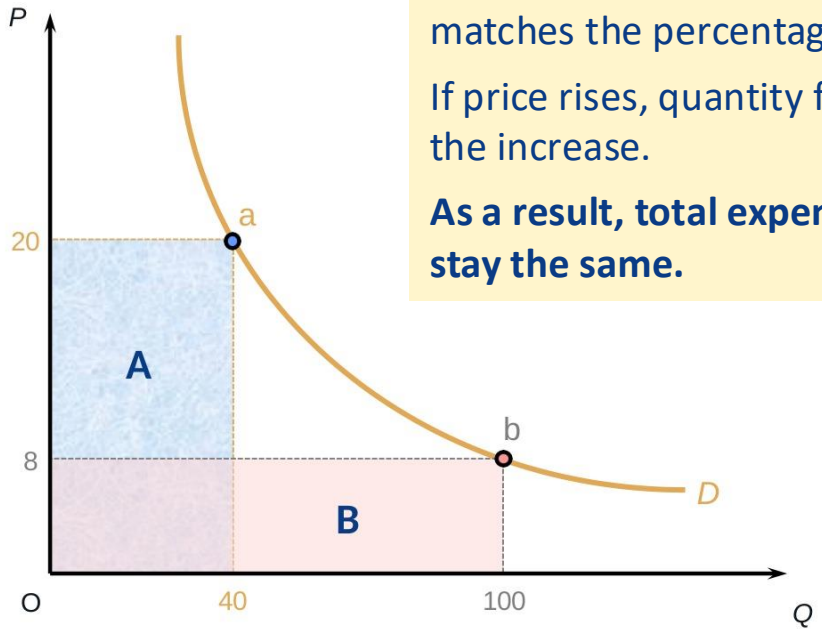
Consumers reduce how much they buy, but only by a relatively small percentage.

So the firm receives more per unit and does not lose very many sales.

Total revenue therefore rises.

Unit Elastic Demand and Expenditure

Unit elastic demand ($P_{\epsilon_D} = 1$)



Unit elastic demand is the middle case.

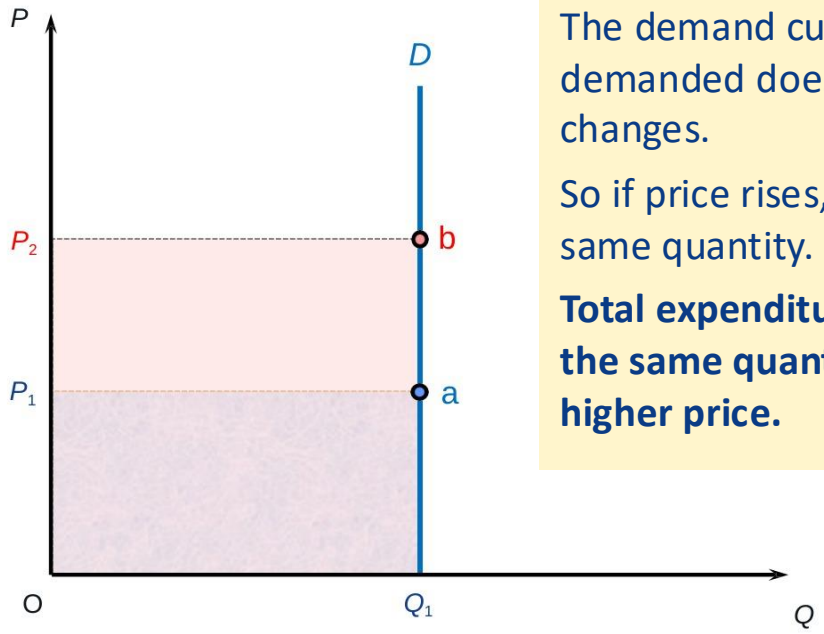
Here the percentage change in quantity exactly matches the percentage change in price.

If price rises, quantity falls by just enough to offset the increase.

As a result, total expenditure and total revenue stay the same.

Perfectly Inelastic Demand

Totally inelastic demand ($P_{\epsilon_D} = 0$)



At the extreme, we have perfectly inelastic demand.

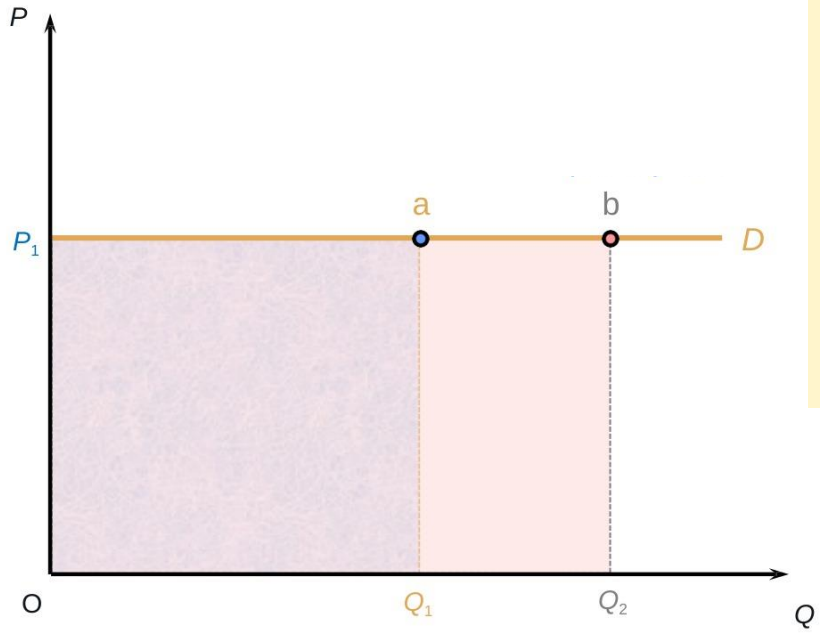
The demand curve is vertical because quantity demanded does not change at all when price changes.

So if price rises, consumers still buy exactly the same quantity.

Total expenditure therefore rises simply because the same quantity is now being purchased at a higher price.

Perfectly Elastic Demand

Infinitely elastic demand ($P_{\epsilon_D} = \infty$)



At the opposite extreme is perfectly elastic demand.

Here the demand curve is horizontal.

At the market price, consumers are willing to buy the quantity offered.

But **if the price rises even slightly above that level, quantity demanded falls to zero.**

Price elasticity of demand and consumer expenditure

- How consumer expenditure(total revenue) changes --- depends on price elasticity of demand:
  - Elastic demand ($P\varepsilon D > 1$) :
    - P rises, Q falls proportionately more, therefore TE falls. ↓TE =P↑*Q↓↓
    - P falls, Q rise proportionately more, therefore TE rises. TE↑=P↓* Q↑↑
  - Inelastic demand ($P\varepsilon D < 1$):
    - P rises, Q falls proportionately less, therefore TE rises TE↑=P↑↑*Q↓
    - P falls, Q rise proportionately less, therefore TE falls ↓TE =P↓↓* Q↑
  - Unit elastic demand ($P\varepsilon D = 1$):
    - P rises(falls), Q falls(rises) with the same proportion, therefore TE stays the same TE=P *Q

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If demand is elastic and a firm lowers its price, what will most likely happen to total revenue?



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0	0	0	0
Increase	Decrease	Stay the same	Cannot be determined

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Show correct answer

Section 1: How Markets Work — Demand and Supply

Other elasticities

Other Demand Elasticities

- The income elasticity of demand

$$\text{Income elasticity} = \frac{\%\,\Delta Q_D}{\%\Delta\, Y}$$

  - Normal goods --- positive income elasticity (income rises, quantity demanded increases)
  - Inferior goods --- negative income elasticity (income rises, quantity demanded decreases)
  - Necessities --- positive but small income elasticity (income rises, quantity demanded increases only slightly)

- The cross-price elasticity of demand

$$\text{Cross-price elasticity } C_{\varepsilon_{D_{AB}}} = \frac{\%\,\Delta Q_{D_A}}{\%\Delta\, P_B}$$

  - Substitutes --- positive cross-price elasticity (Price of B rises, people will buy more A instead)
  - Complements--- negative cross-price elasticity (Price of B rises, quantity demanded for A reduce)

Price Elasticity of Supply

- Price elasticity of supply looks at the seller's side of the market.
  - Measures how responsive quantity supplied is to a change in price

$$P_{\varepsilon_S} = \frac{\%\Delta Q_S}{\%\Delta P}$$

  - Sign: positive (positive relation between price and quantity supplied)
  - **Elastic supply** (P_{ε_S} >1) quantity supplied changes by a larger percentage than price; **inelastic supply** (P_{ε_S} < 1) quantity supplied changes by a smaller percentage than price; **unit elastic supply** (P_{ε_S} =1) the percentage changes are the same.
  - Determinants of price elasticity of supply: **Time** --- supply is more elastic in the long run because producers have more time to adjust production.
  - Measurement
    - Arc method (midpoint method to calculate percentage) $P_{\varepsilon_S} = \frac{(Q_2 - Q_1)/[\frac{Q_2+Q_1}{2}]}{(P_2 - P_1)/[\frac{P_2+P_1}{2}]}$
    - Point elasticity $P_{\varepsilon_S} = \frac{dQ_S}{dP} * \frac{P}{Q}$

Section 2: Background to Demand

2.1 Consumer Choices — Utility

Consumer Choice Explains Demand



£10

How to spend?







Consumer choice explains demand.



Consumer Choice and Utility

- **Utility** is the satisfaction a consumer gets from consuming a good or service.

- Utility theory: rational consumers make choices to **maximize their utility.**

- **Total utility (TU)** is the total satisfaction a person gains from all those units of a commodity consumed within a given time period.

$$TU = U(Q)$$

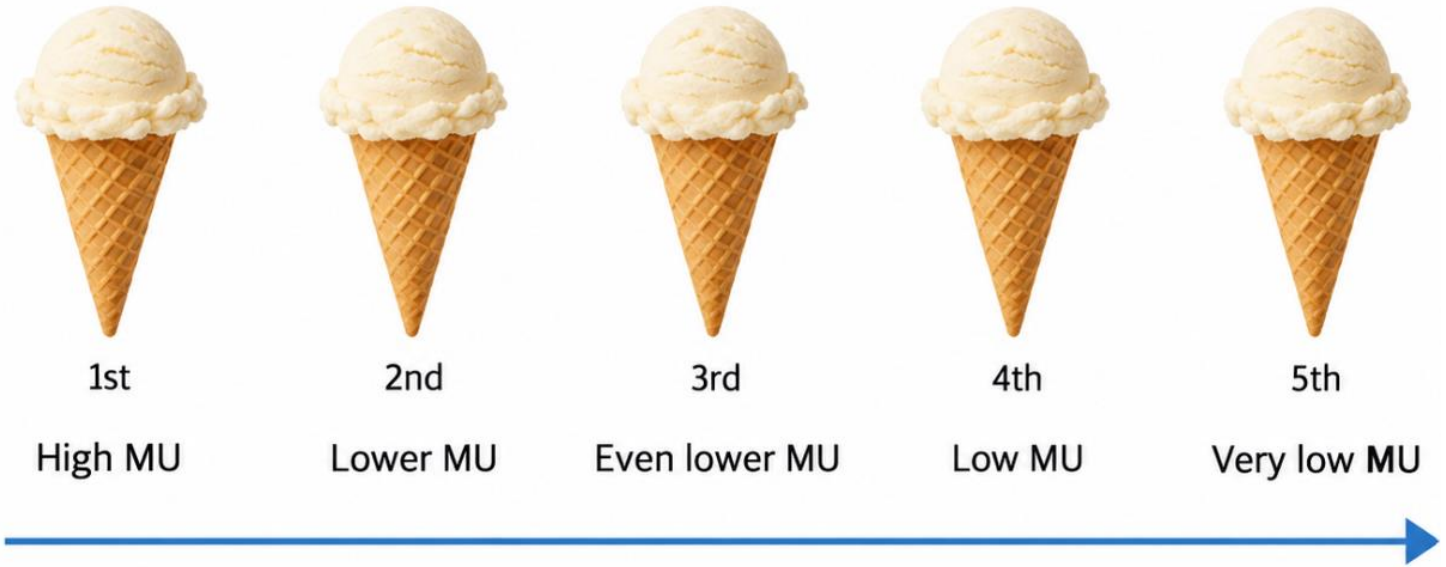
- **Marginal utility (MU)** is the additional satisfaction gained from consuming one extra unit within a given period of time.

$$MU = \frac{dTU}{dQ}$$



Diminishing Marginal Utility

Principle of diminishing marginal utility: The more of a product a person consumes, the less will be the additional utility gained from one more unit.



The more units of the same product you consume, the less additional satisfaction you normally receive from one more unit.

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Imagine you are very hungry and eat slices of pizza. Which slice is likely to give you the greatest additional satisfaction?



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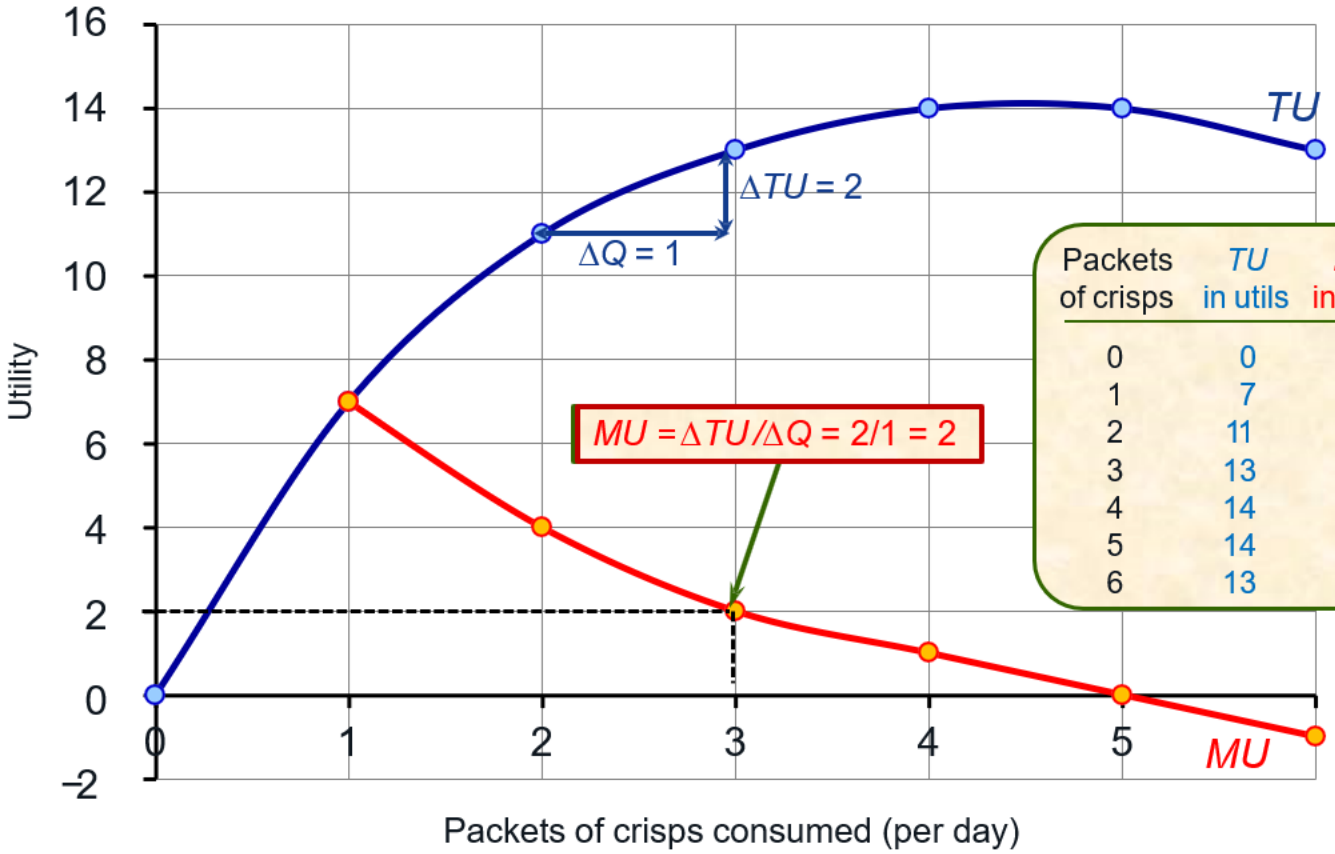
0

0	0	0	0
First slice	Third slice	Fifth slice	Every slice gives the same additional satisfaction

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Show correct answer

Example: Utility from Consuming Crisps



Packets of crisps	TU in utils	MU in utils
0	0	-
1	7	7
2	11	4
3	13	2
4	14	1
5	14	0
6	13	–1

If you have already eaten far too much, consuming another unit may actually reduce your satisfaction.

Utility, Willingness to Pay and Consumer Surplus

- **Consumer choice depends on:**
  - utility;
  - income or budget;
  - the prices of goods and services.
- **Utility** represents the satisfaction a consumer receives from consuming a good or service. **Utility is abstract** → we need a way to measure it.
- **Willingness to pay (WTP)** is the maximum amount a consumer is prepared to pay for a good or service → *i.e. how much the satisfaction is worth.*
- *Willingness to pay (WTP) = monetary value of utility.*
- **Consumer surplus** is the difference between the amount a consumer is willing to pay and the amount they actually pay.

Optimal Consumption: One-Good Case

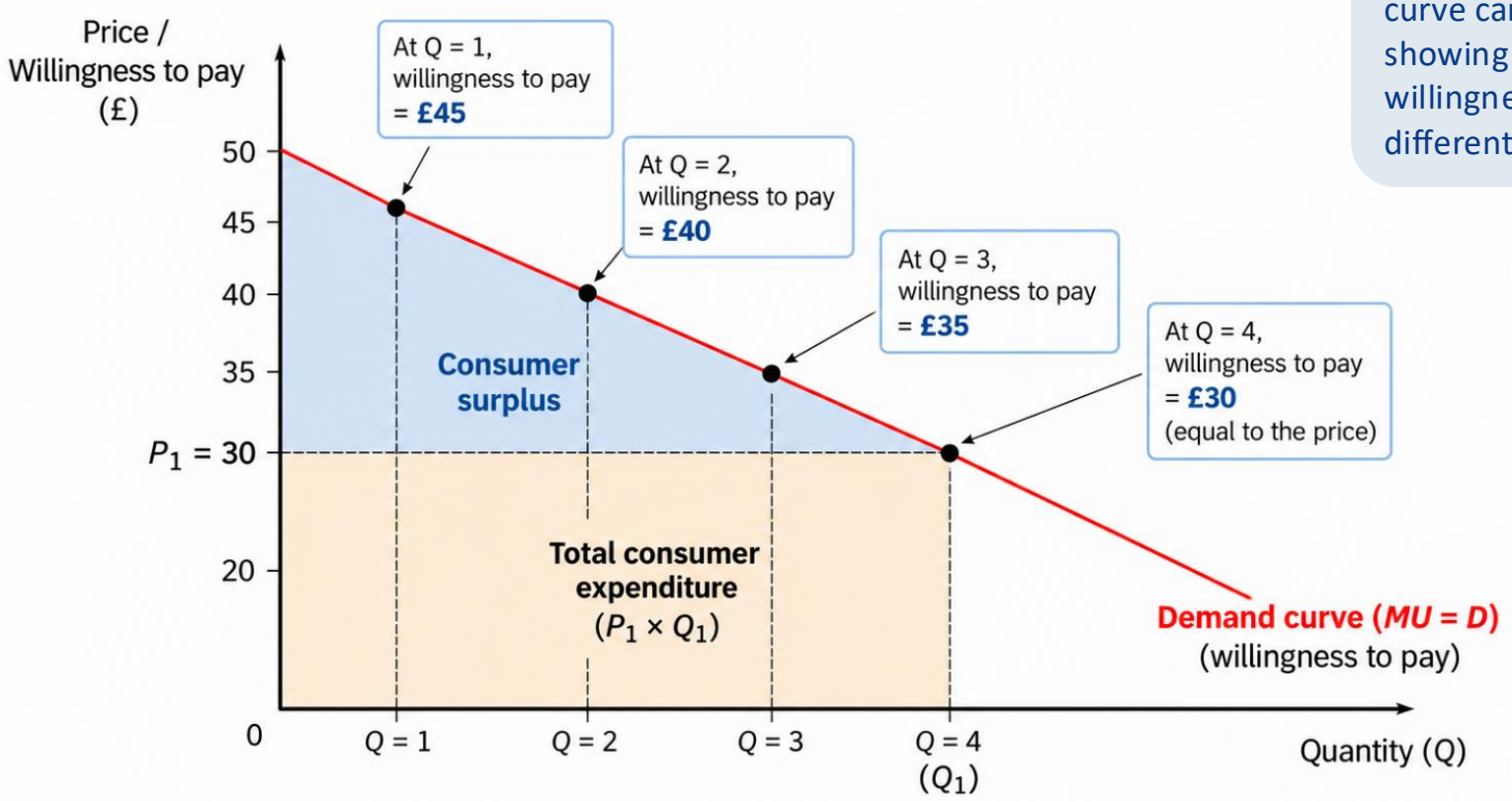
- **Marginal consumer surplus (MCS)** is the difference between what you are willing to pay for one more unit of a good and what you are actually charged.
  - It is computed as: $MCS = MU - P$
    where MU = marginal utility, expressed in monetary terms, and P = price paid

- **Total consumer surplus (TCS)** is the sum of the marginal consumer surplus from all units consumed.
  - It is computed as $TCS = TU - TE$
    where TU = total utility, expressed in monetary terms, and TE = total expenditure

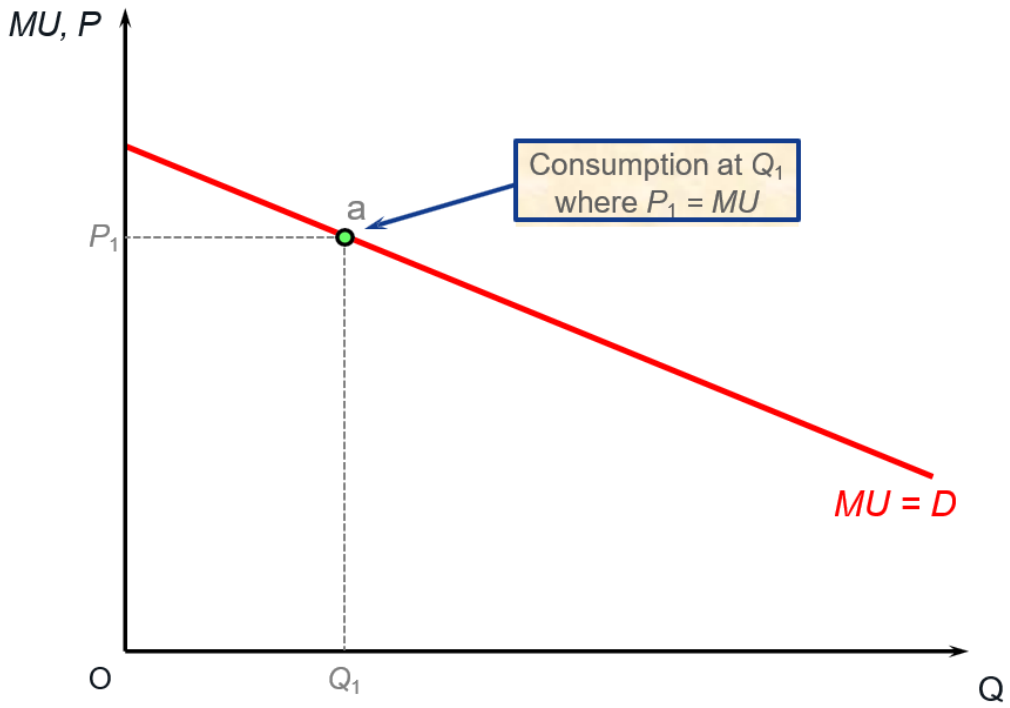
- *Rational consumer behaviour: The attempt to maximize total consumer surplus.*

Measuring Consumer Surplus Using the Demand Curve

Remember that a demand curve can be interpreted as showing consumers' willingness to pay for different quantities.



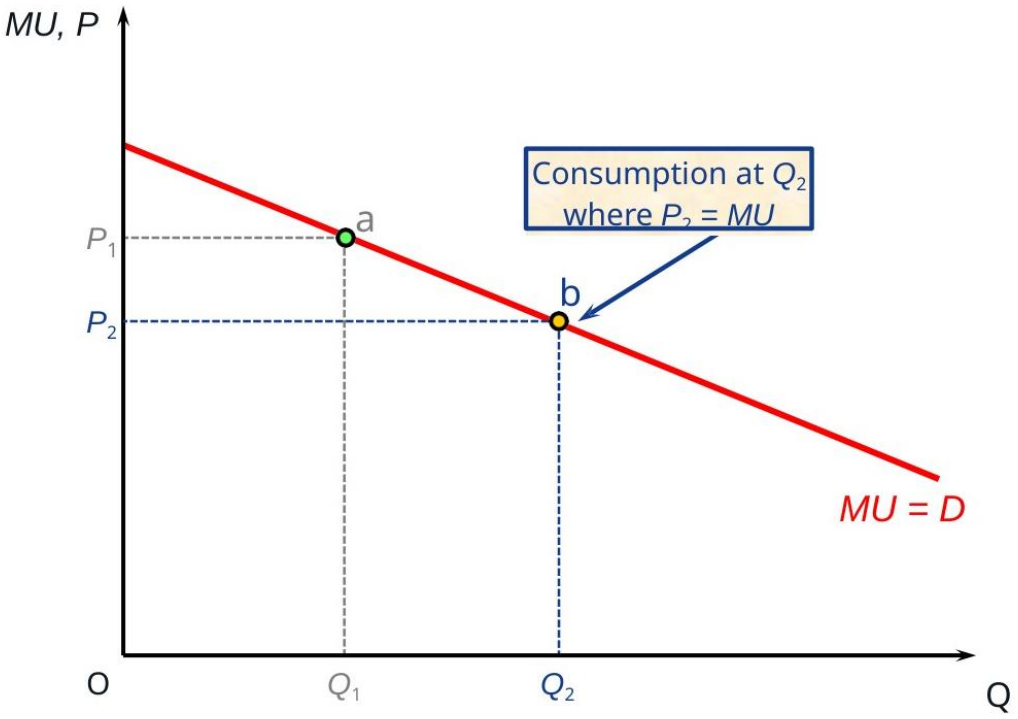
Consumer Surplus at Q1



At Q1:

- The demand curve shows the consumer's willingness to pay for the next unit.
- At Q1, WTP = MU = P1, so the final unit is worth exactly what it costs.
- Therefore, the marginal consumer surplus of the final unit is zero.
- For all earlier units, WTP > P1, so they generate positive consumer surplus.
- This is why Q1 is the optimal quantity at price P1.

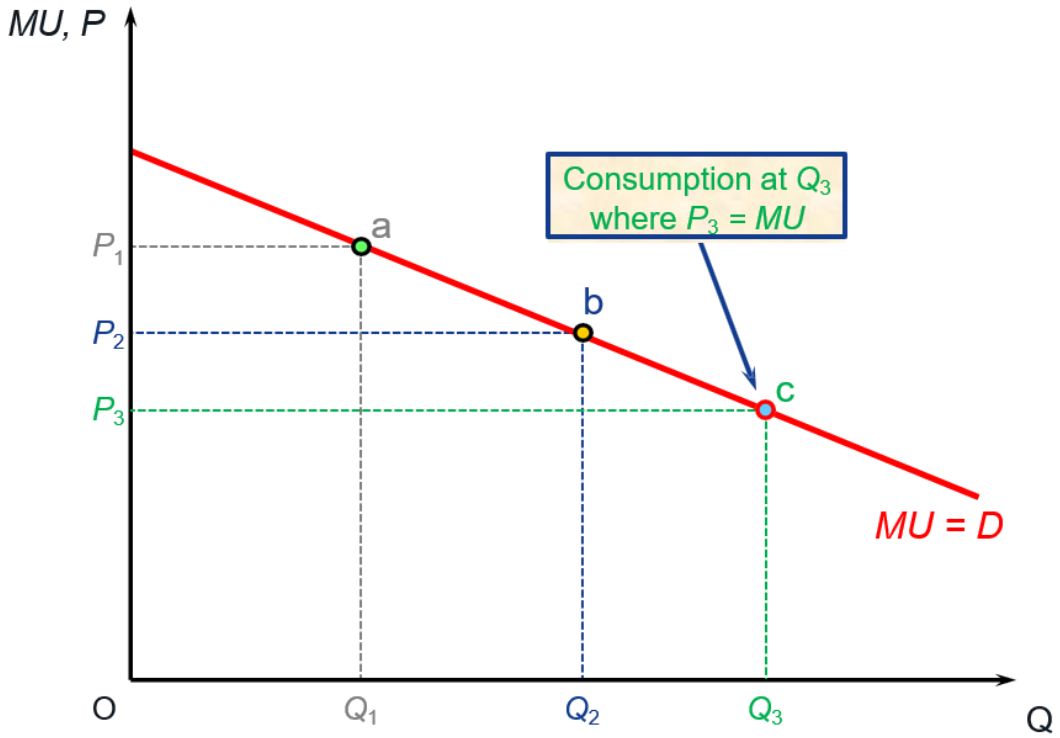
What Happens if Consumption Increases to Q2?



When price falls from P1 to P2:

- At the lower price P2, buying only Q1 is no longer optimal, because at Q1 the consumer's willingness to pay is still above the new price.
- So the consumer has an incentive to buy more.
- As consumption increases, marginal utility (WTP) falls.
- Consumption rises until Q2, where WTP = MU = P2.
- So Q2 is the new optimal quantity at the lower price.

Why Q3 Is Too Much Consumption



Now suppose the market price remains at P2, but the consumer goes beyond the optimal quantity Q2 and consumes Q3:

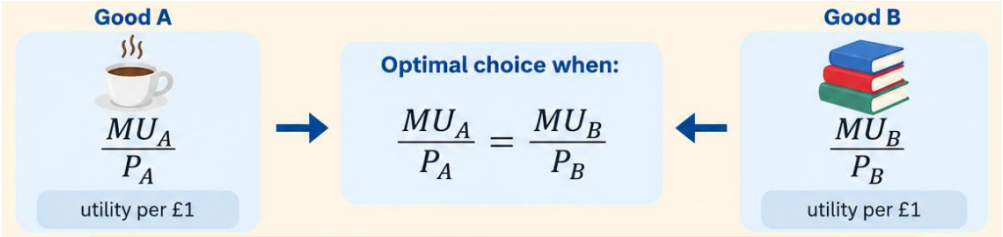
- Look at the marginal utility at Q3. It has fallen to P3.
- But the consumer still has to pay the market price P2.
- So at Q3, the additional satisfaction from the last unit is worth less than the amount the consumer has to pay for it.
- In other words, MU < P and marginal consumer surplus is negative.
- This is why Q3 is too much consumption at the price P2.

From One Good to Choosing Between Goods

- **Why the one-good model is limited**
  - **The value of spending an extra £1 can change.**
    It depends on how much income is left and what else we could buy with that £1.
  - **Utility from one good may depend on other goods.**
    Example: coffee may give more satisfaction when consumed with milk or sugar.



- **Choosing the optimal combination of goods**
  - **Equi-marginal principle**: Consumers maximise total utility by allocating income so that marginal utility per £1 spent is equal across goods.



If one good gives more extra utility per £1, spend more on that good until the ratios are equal.

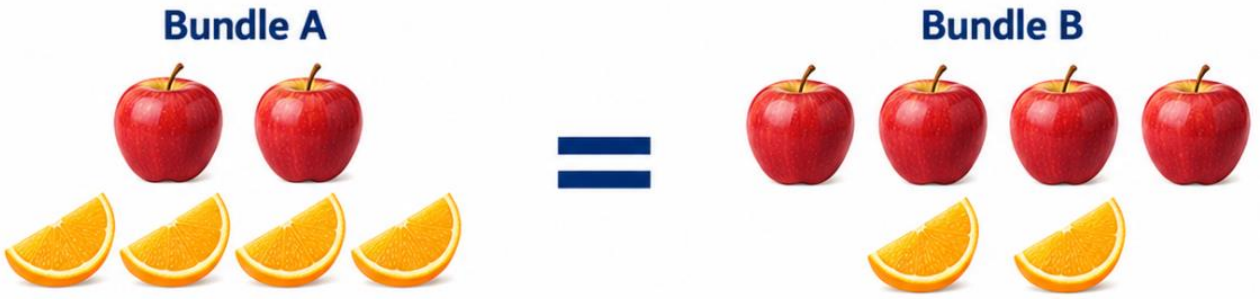
Section 2: Background to Demand

2.2 Consumer Choices — Indifference Analysis

Indifference Curves: What the Consumer Wants

- **Indifference analysis**
  - does not measure utility in numerical units
  - compares different combinations, or bundles, of goods
  - ranks these bundles according to the consumer's preferences

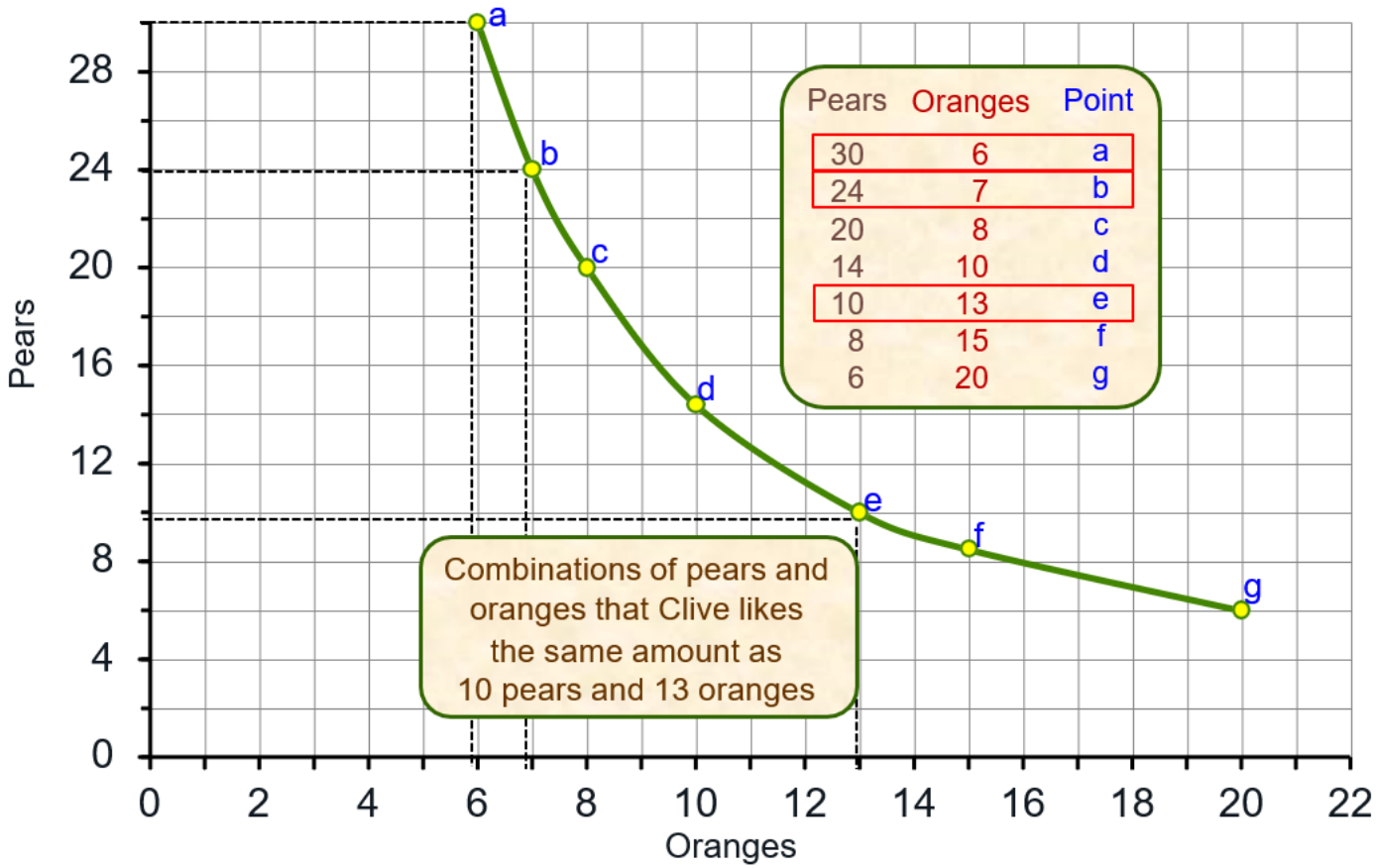
- **Indifference curve**
  - shows all combinations of two goods that give the consumer **the same level of satisfaction**; the consumer is indifferent between any two points on the same curve



Different bundles can give the same level of satisfaction.

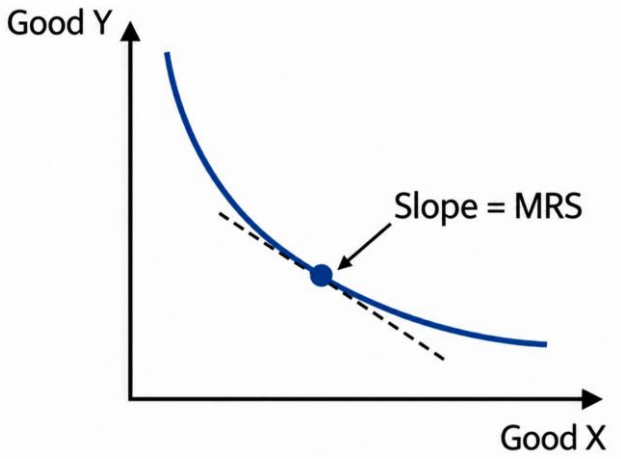
Indifference Curve

Shows all the various combinations of two goods that give an equal amount of satisfaction or utility to a consumer

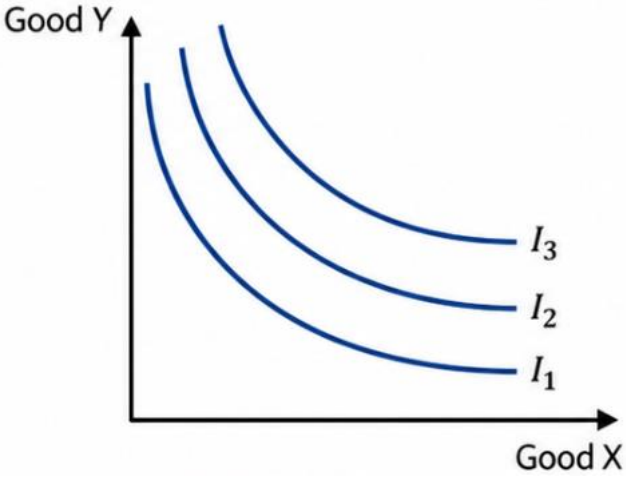


Properties of Indifference Curves

- **Marginal rate of substitution (MRS)**
  - the slope of indifference curve = the rate at which the consumer is willing to substitute one good for the other
  - usually diminishes as the consumer moves along an indifference curve

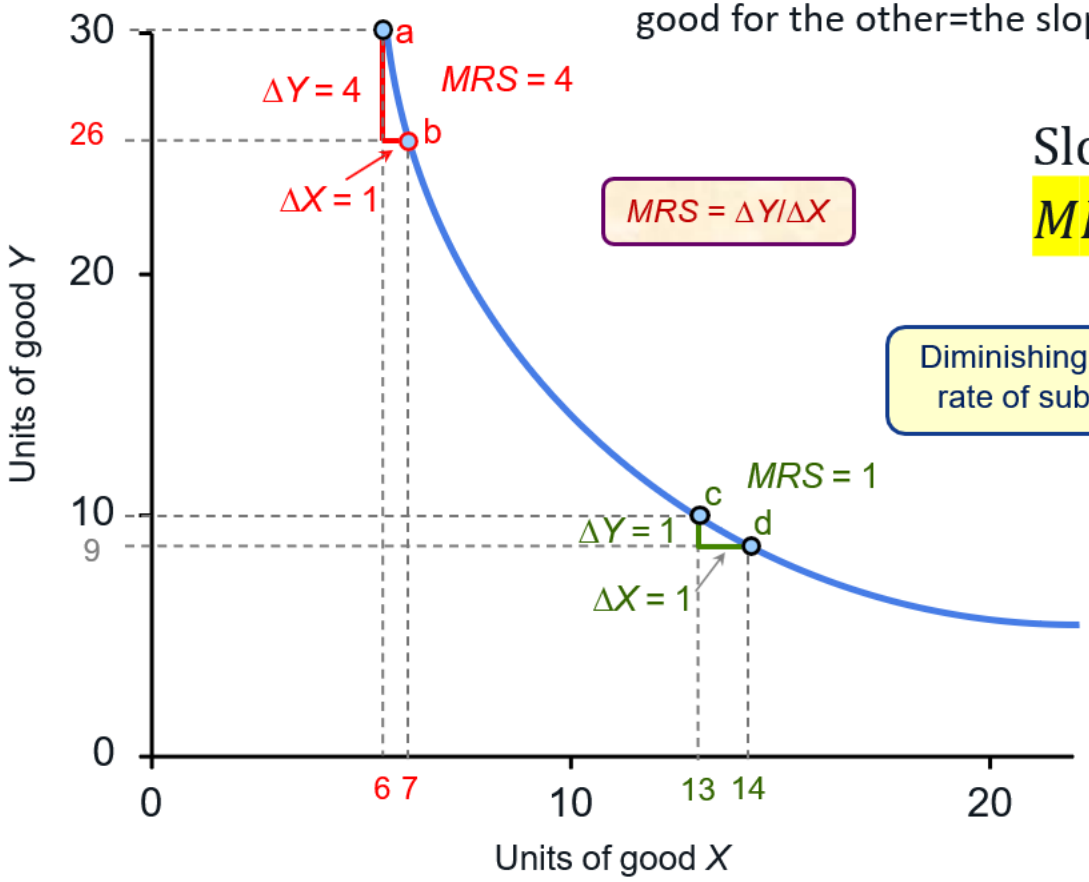


- **Indifference map**
  - contains several indifference curves
  - higher indifference curves represent higher levels of satisfaction
  - two indifference curves cannot intersect each other



Marginal Rate of Substitution

the rate at which the consumer is willing to substitute one good for the other=the slope of indifference curve



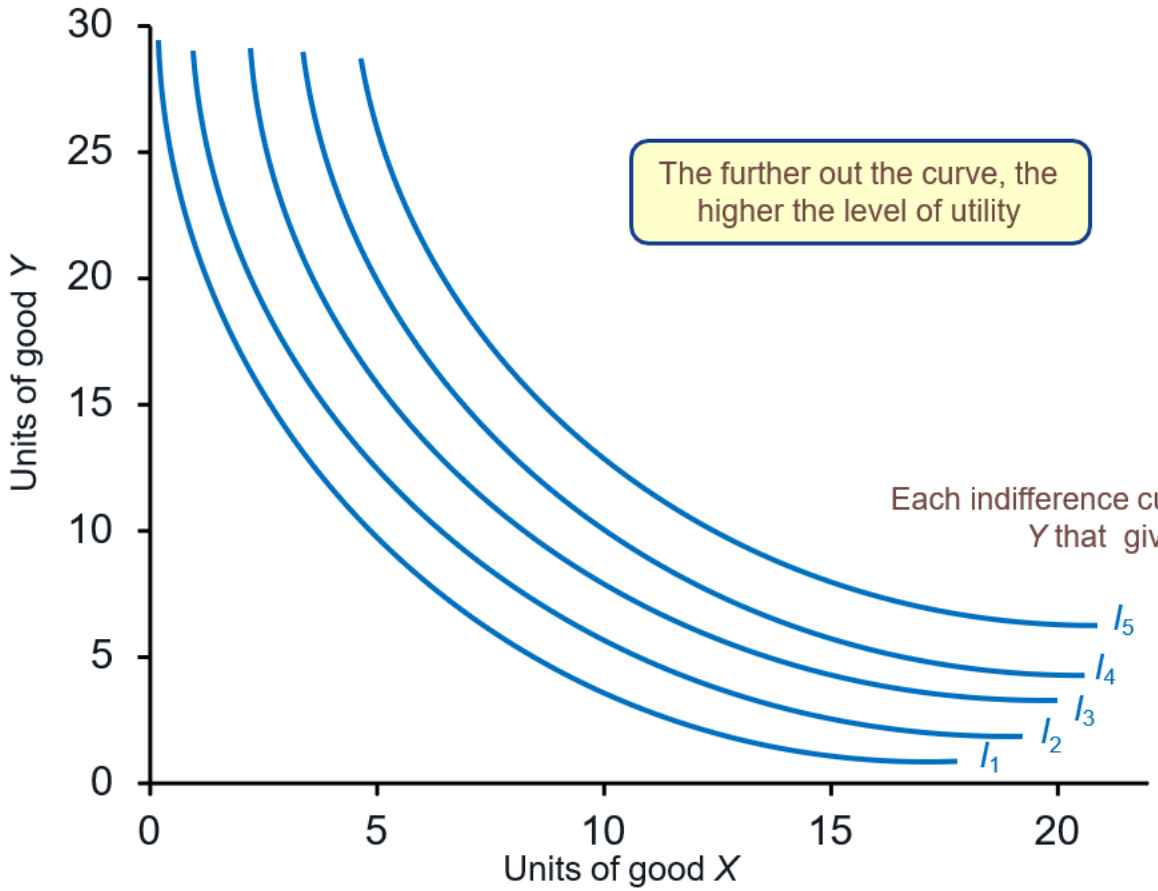
MRS = ΔY/ΔX

Slope :

$MRS = MU_X / MU_Y$

Diminishing marginal rate of substitution

Indifference Map

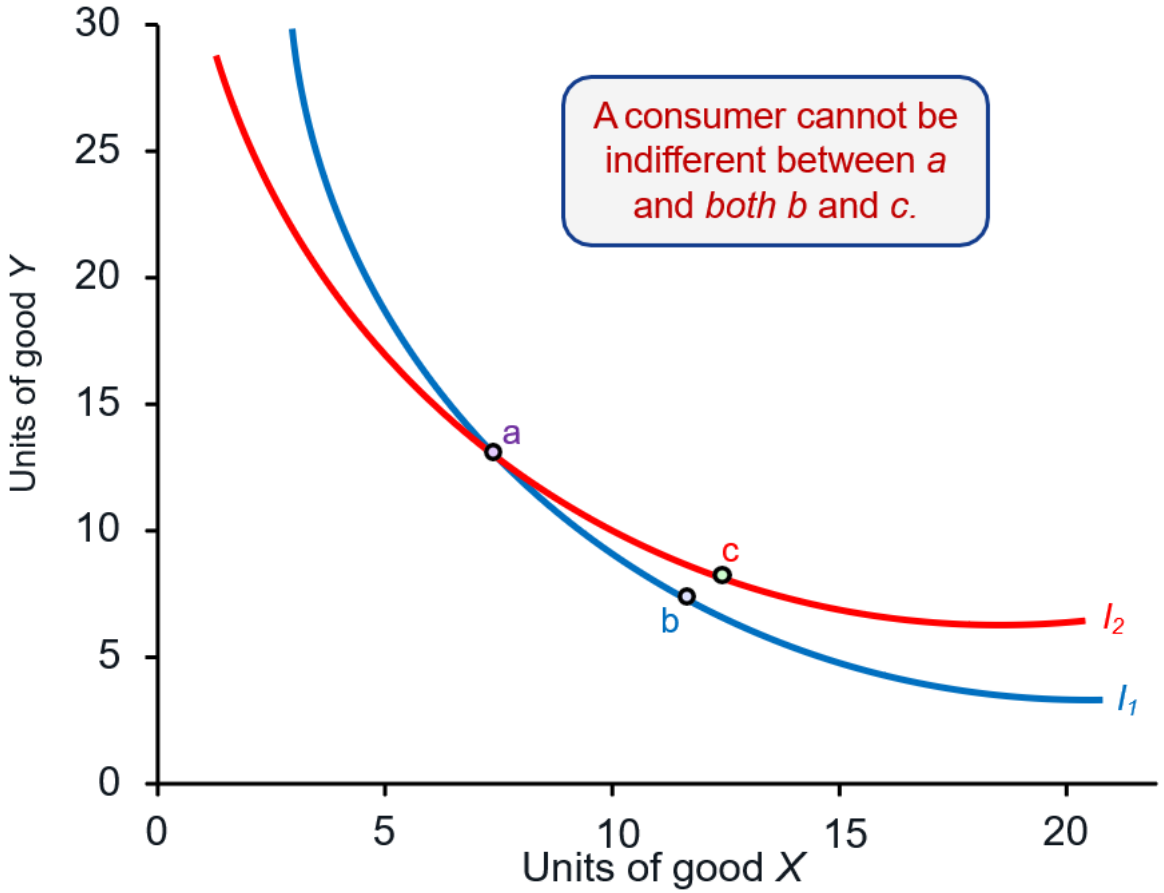


The further out the curve, the higher the level of utility

Each indifference curve shows all combinations of *X* and *Y* that give a particular level of utility.

Of course, consumers cannot simply choose the highest curve they want.
Their income and prices limit what they can actually afford.

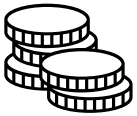
Why Indifference Curves Cannot Cross



Budget constraints: What a Consumer Can Afford

- **Budget line**
  - Shows all combinations of two goods that a consumer can afford, given their income and the prices of the two goods.

- **Changes in the budget line**
  - a change in income leads to a parallel shift in the budget line

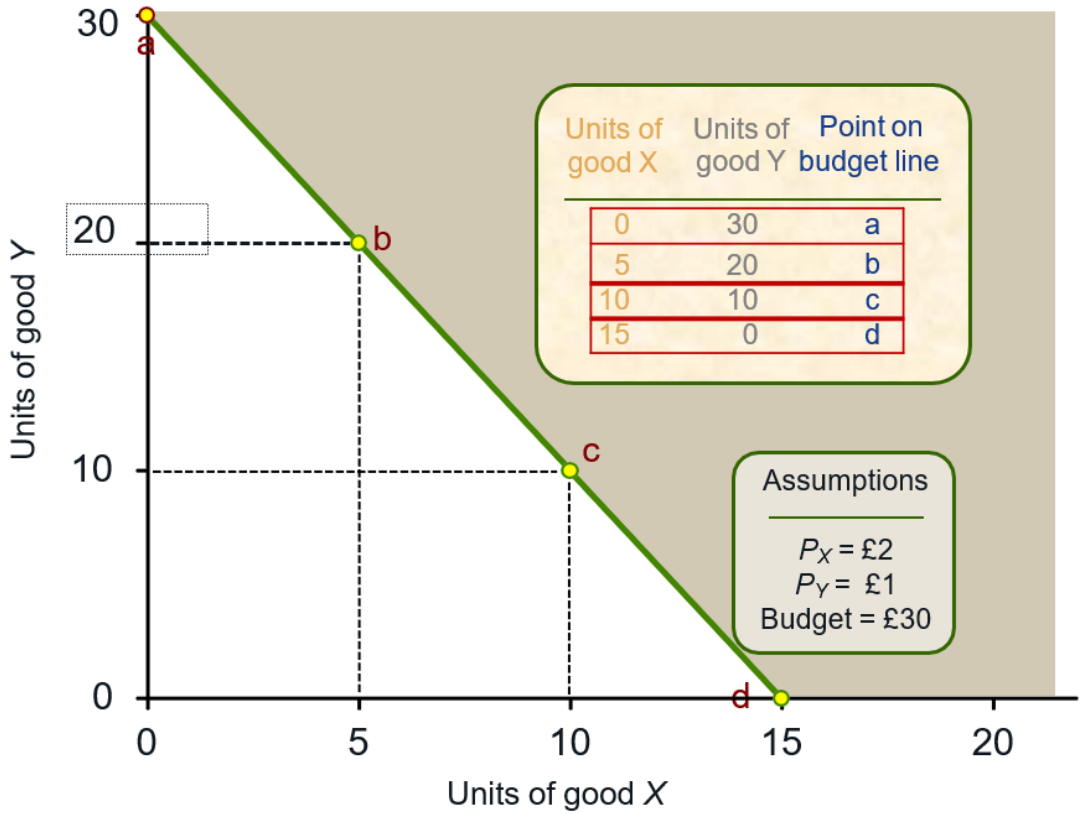


  - a change in the price of one good causes the budget line to rotate.



The Budget Line

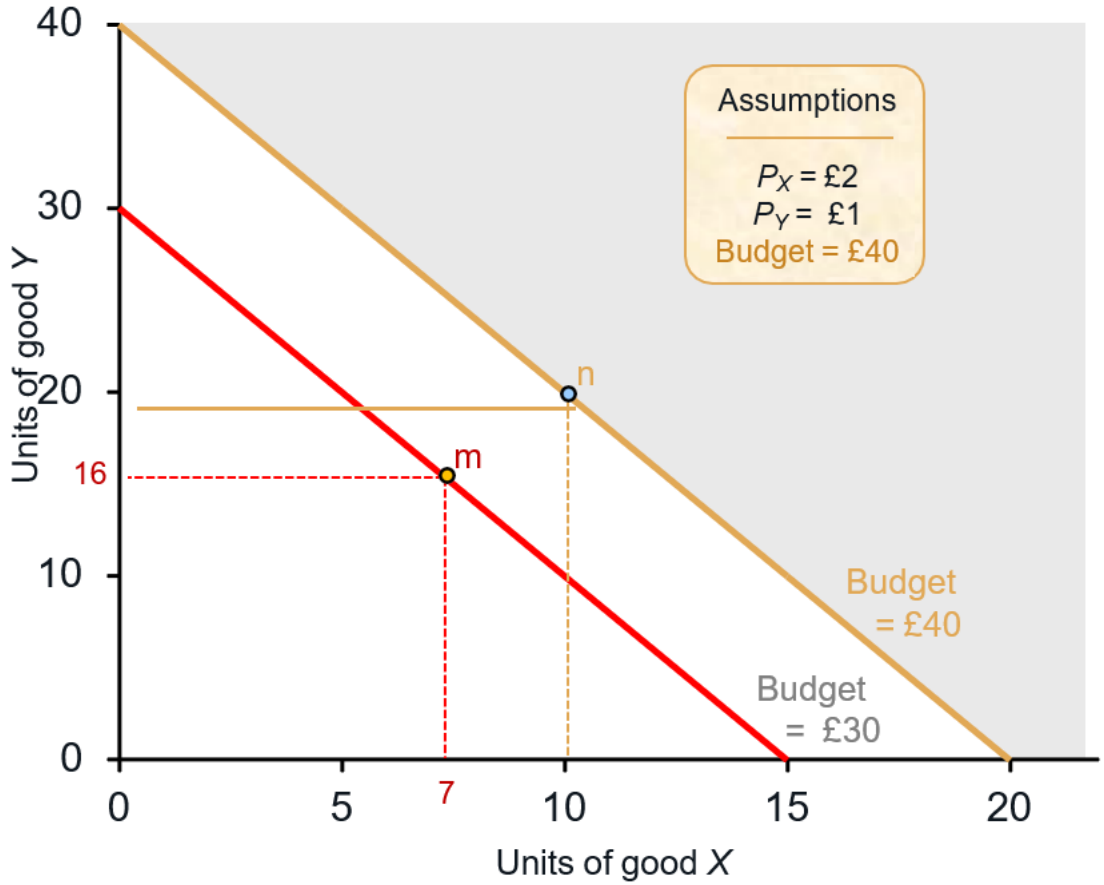
Budget line A graph showing all the possible combinations of two goods that can be purchased at given prices and for a given budget.



Slope of budget line
= 30/15 = P_X /P_Y

The budget line represents the boundary of what the consumer can afford.

Effect of an Income Increase on the Budget Line



Slope of budget line
= 30/15 = Px /Py

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If a consumer's income rises while the prices of both goods remain unchanged, what happens to the budget line?

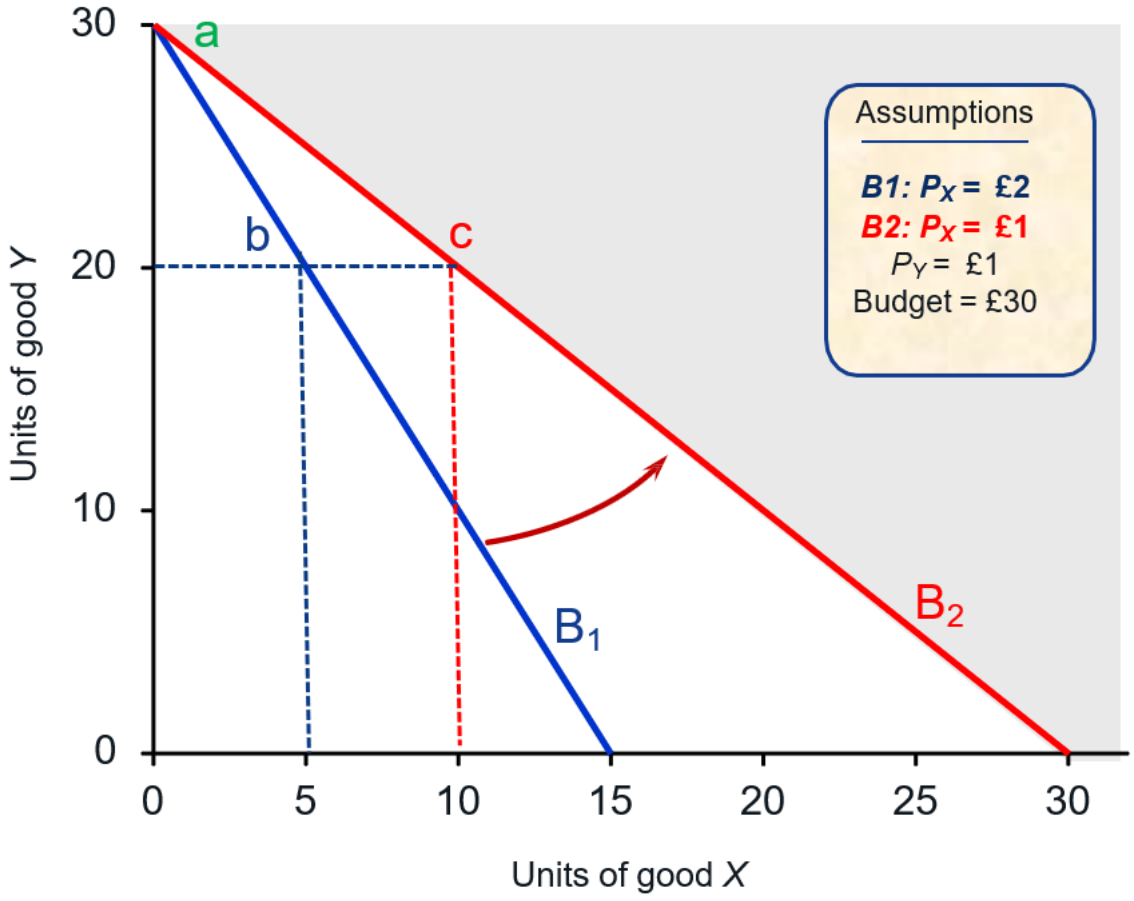
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0	0	0	0
Shifts outward in parallel	Shifts inward in parallel	Rotates inward	Does not change

0 of 66 responded

Show correct answer

Effect of a Fall in the Price of Good X



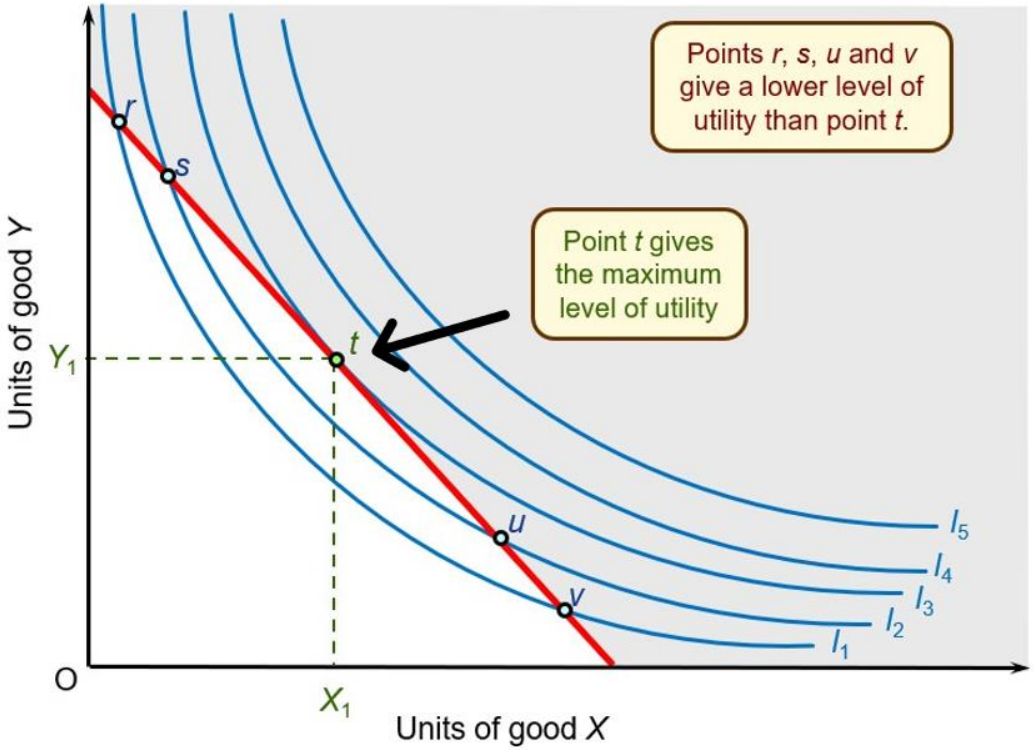
The consumer can afford more X than before.

But nothing has changed about the amount of Y they can buy if they spend all their income on Y, because the price of Y has not changed.

Optimal Consumer Choice

- **Consumer choice depends on two things:**
  - the consumer's budget constraint;
  - the consumer's preferences.
- The consumer chooses the highest indifference curve that can be reached given the budget constraint.
- At an interior optimum, the indifference curve is tangent to the budget line.

The Optimal Consumption Point



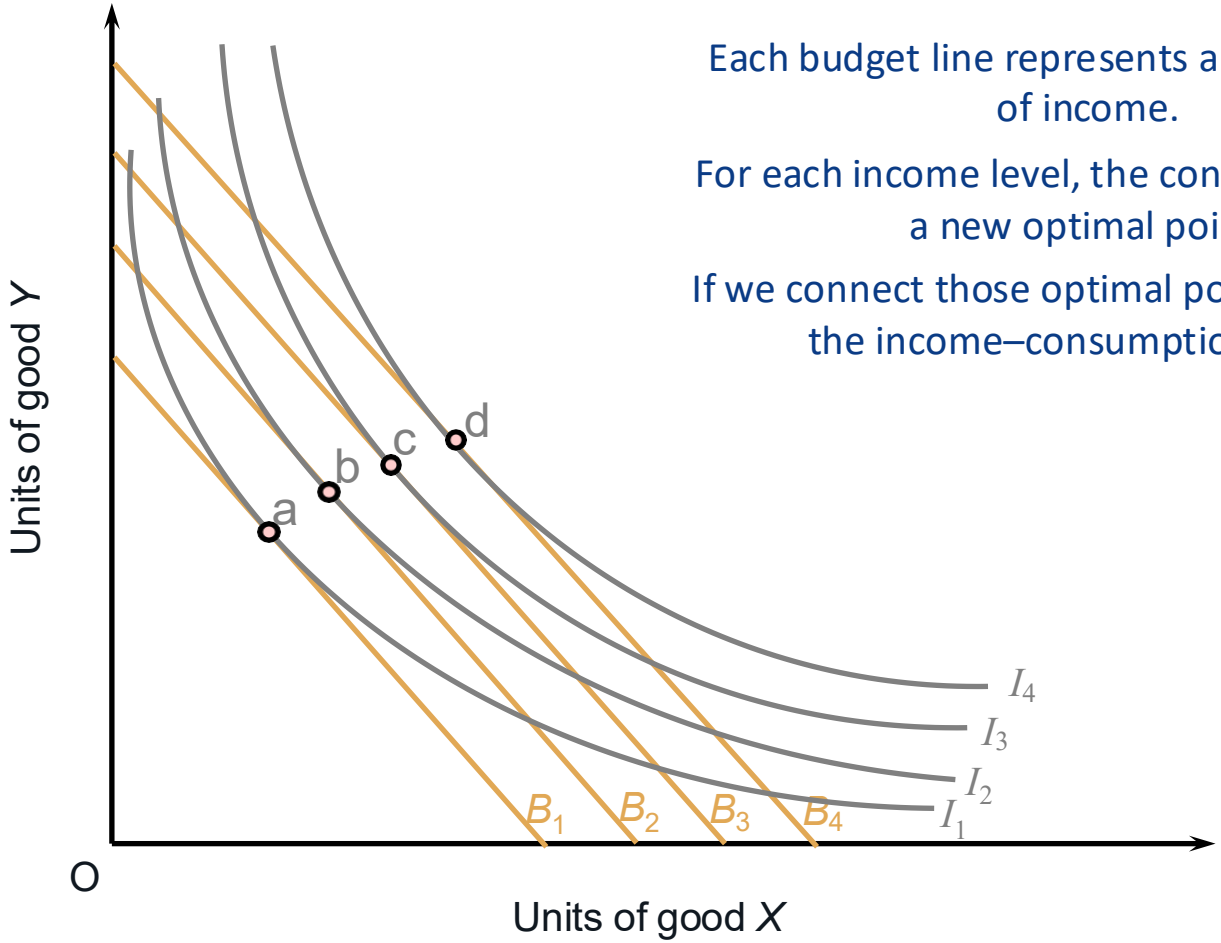
Equating the marginal rate of substitution with the price ratio:

$$MRS = MU_X/MU_Y$$
$$= P_X/P_Y$$

How Income Changes Consumer Choice

- The **income–consumption curve** shows how the consumer's chosen combination of goods changes as income changes.
- Increase in Income
  —> parallel shift in the budget lines (right)
    —> normal goods (Q consumed increased)
    —> inferior goods (Q consumed decreased)

Income–Consumption Curve

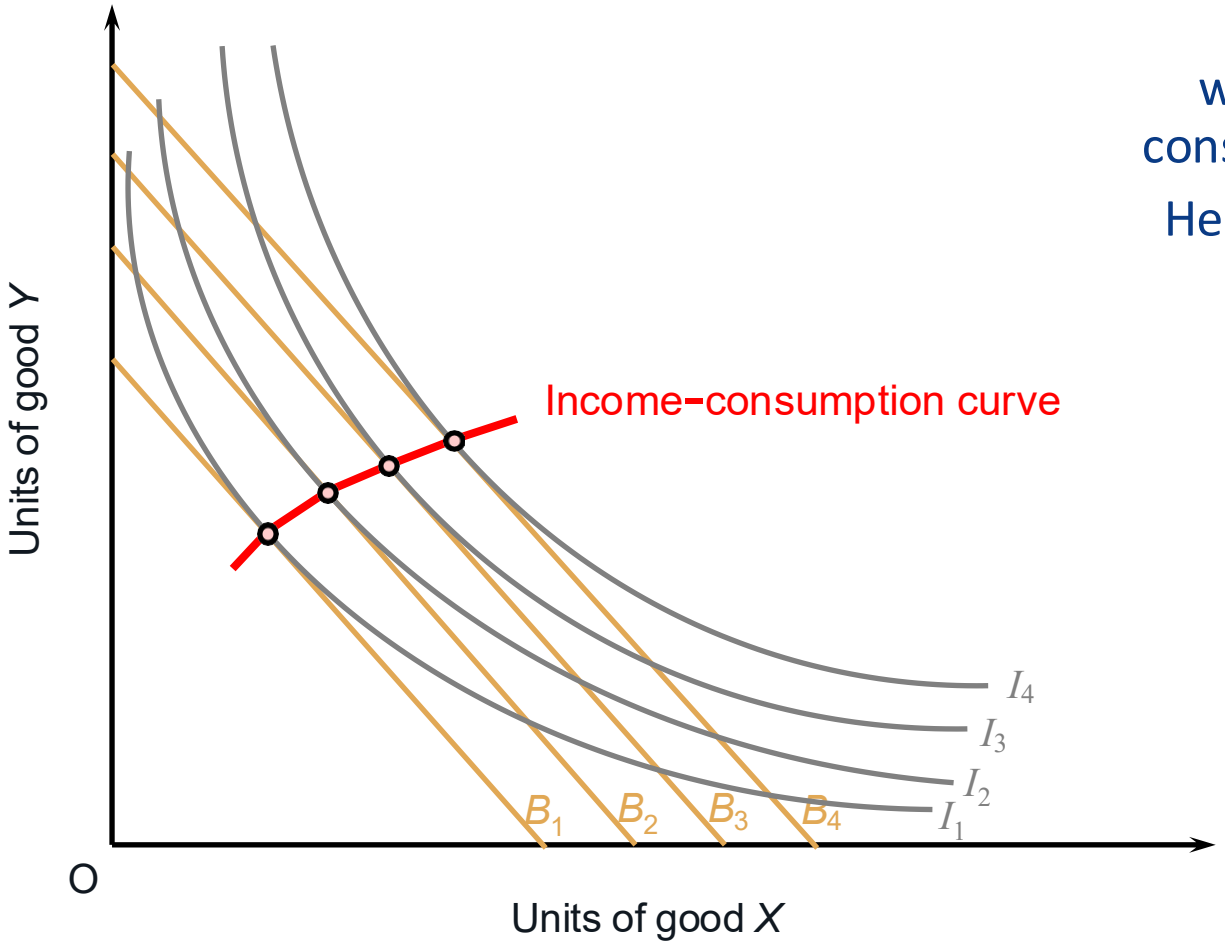


Each budget line represents a different level of income.

For each income level, the consumer chooses a new optimal point.

If we connect those optimal points, we obtain the income–consumption curve.

Normal Goods and the Income-Consumption Curve

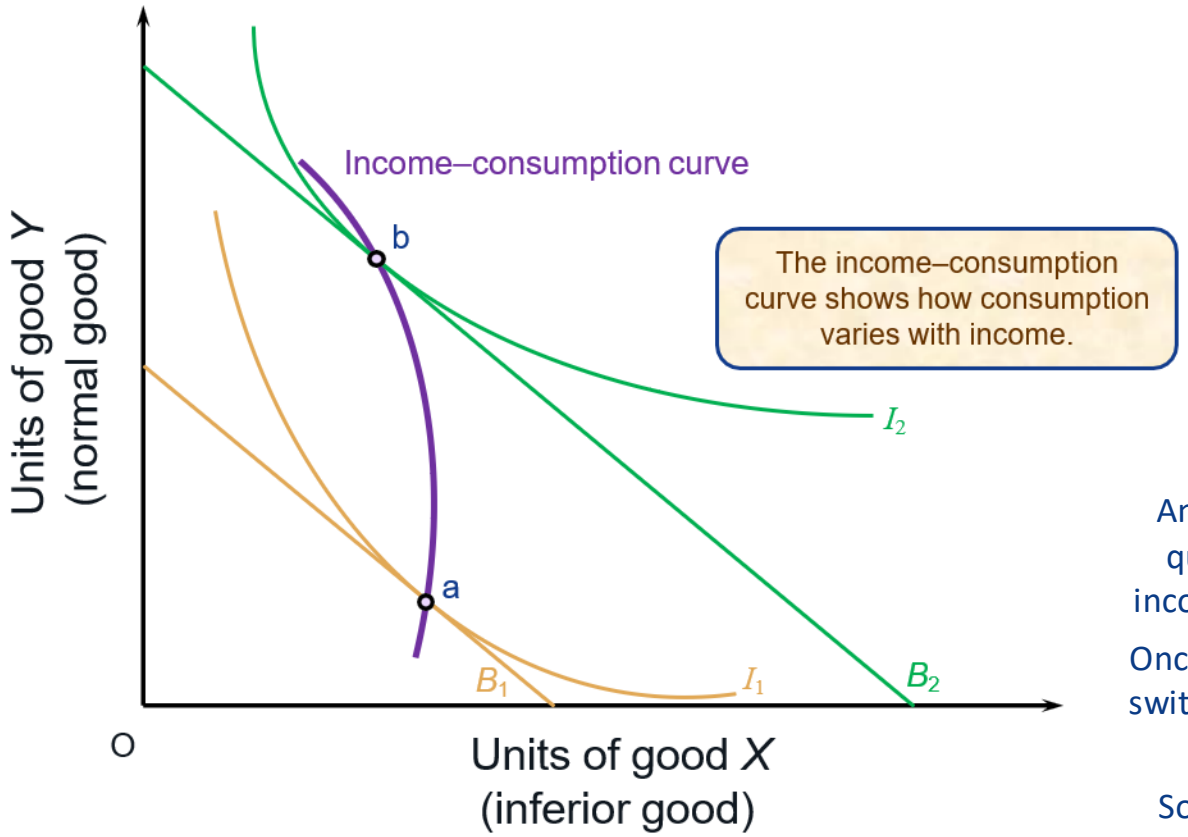


Normal goods:

when income increases, consumers want to buy more

Here both goods behave as normal goods.

Inferior Goods and the Income-Consumption Curve



An inferior good does not mean a bad-quality good. It simply means that, as income rises, consumers choose less of it.

Once people have more income, they may switch from a cheaper alternative towards something they prefer.

So as the budget expands, the optimal quantity of the inferior good can actually fall.

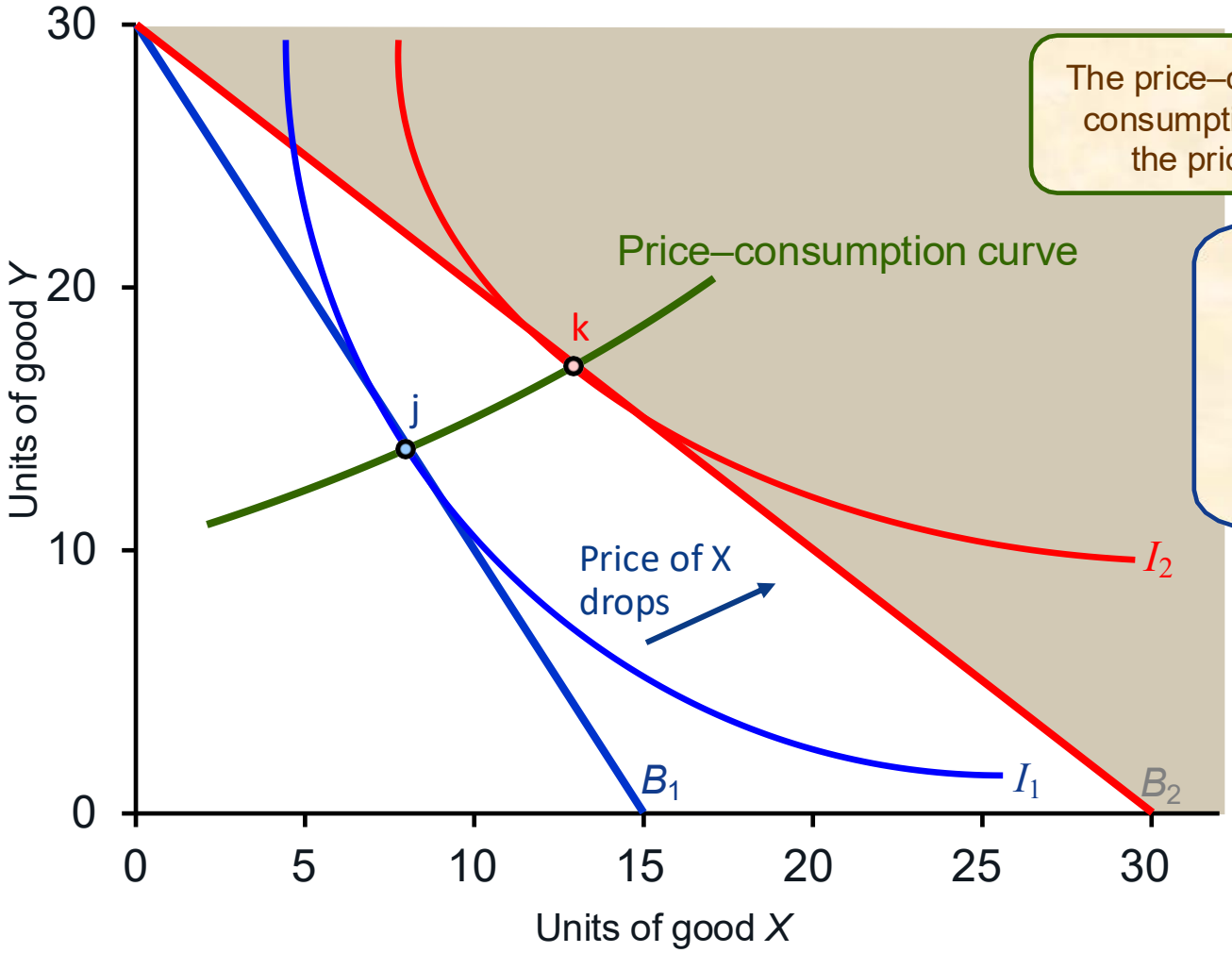
How a Price Change Affects Consumer Choice

- The **price–consumption curve** shows how the consumer's chosen combination of goods changes when the price of one good changes.

  Increase in price of a good (P1) ↑ — budget line rotates inward — quantity consumed falls (Q1) ↓

- The resulting relationship between the price of the good and the quantity demanded can be used to derive the **demand curve**.
- The total effect of a price change can be divided into:
  - **a substitution effect;**
  - **an income effect.**

Effect of a Fall in the Price of Good X



The price–consumption curve shows how consumption is affected by a change in the price of one of the two goods

Assumptions

Initial price of X ***B1:*** P_X = **£2**

New price of X: ***B2:*** P_X = **£1**

Price of Y: P_Y = £1

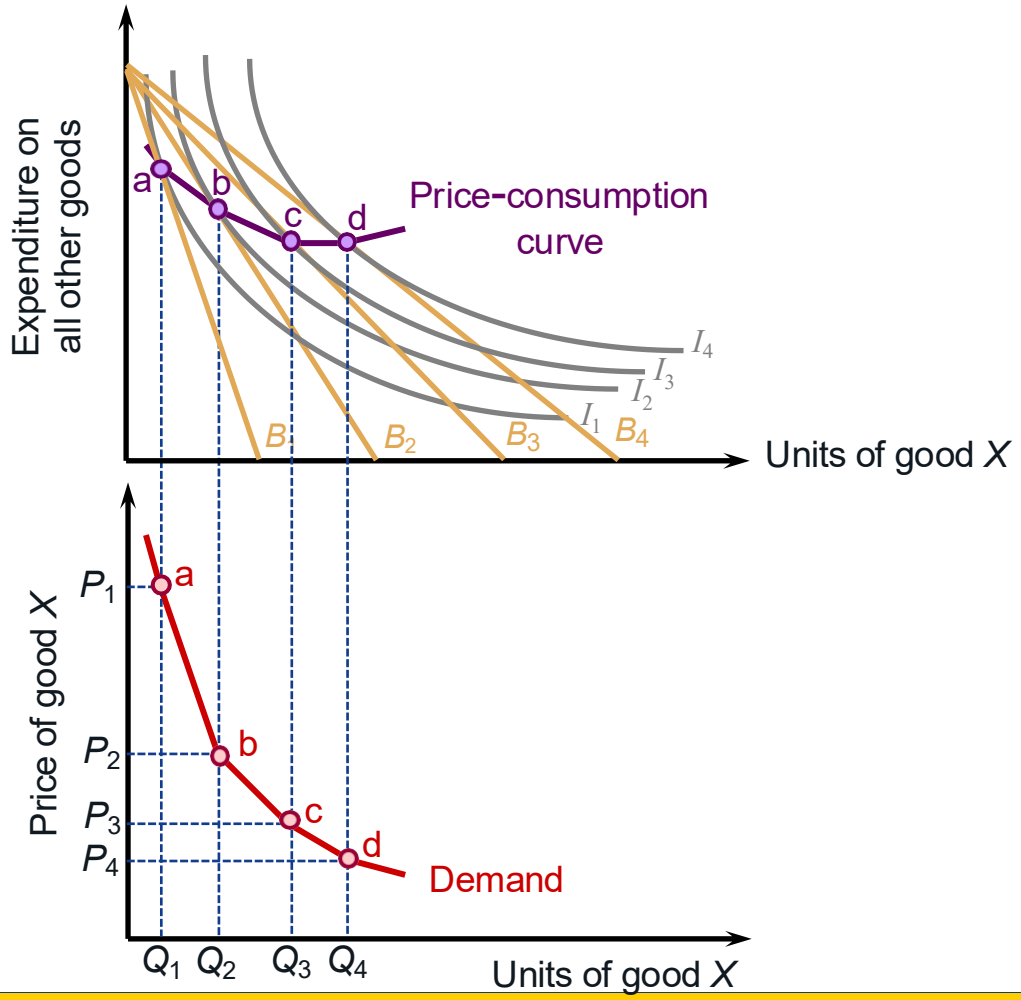
Income: £30

Price-consumption Curve:

Upward-sloping curve → X and Y may be complements

Downward-sloping curve → X and Y may be substitutes

Deriving the Demand Curve from the Price–Consumption Curve



The price-consumption curve illustrates how changes in price (P) affect changes in consumption (Q_D), holding income and other prices constant.

Each optimal choice gives us:

- a price of good X;
- the corresponding quantity demanded of good X.

Plotting these combinations of price and quantity gives the **demand curve**.

Income and Substitution Effects

Imagine one good, X, becomes more expensive. Then two different things happen at the same time.

1. Substitution effect 

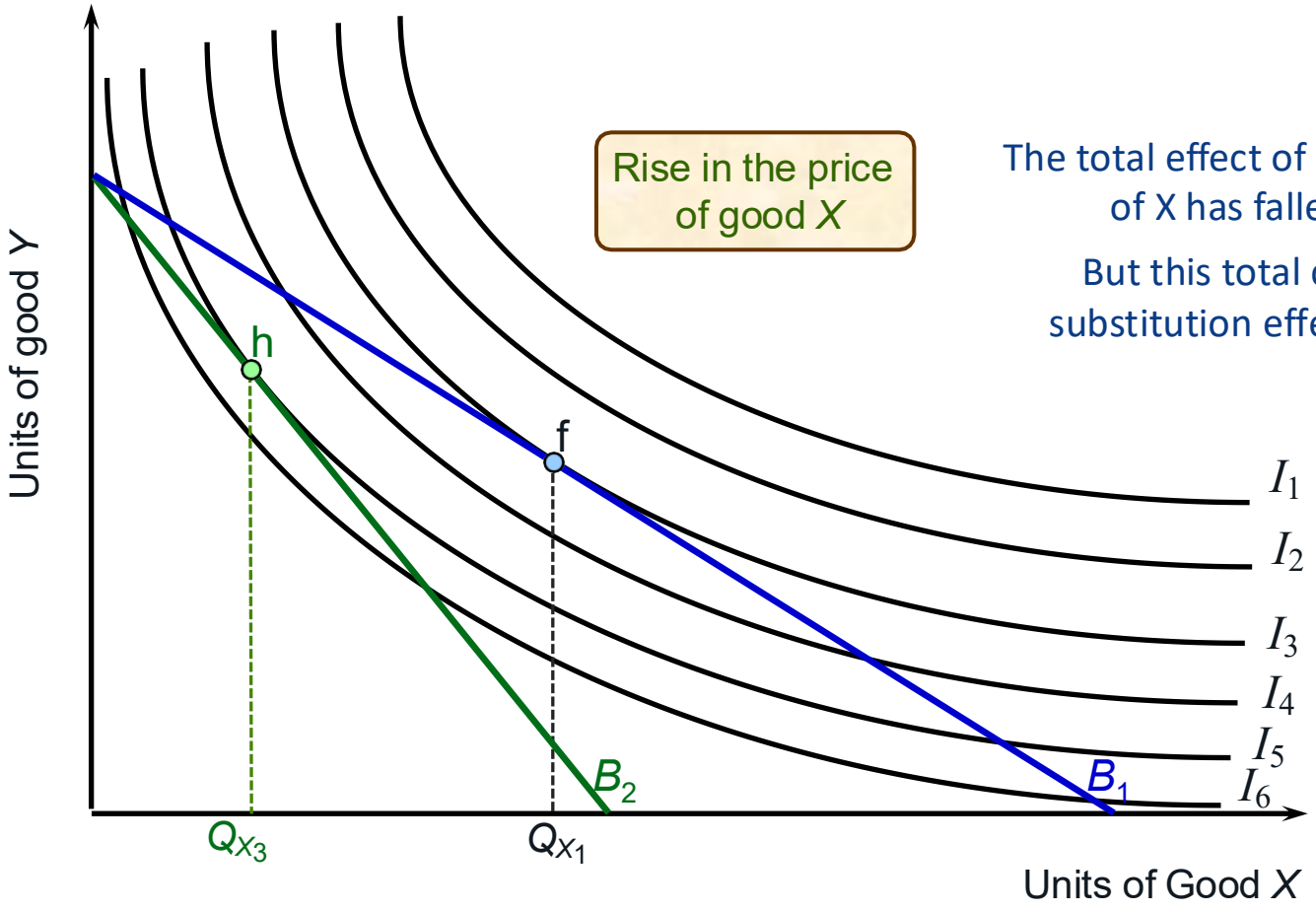
- If good X becomes more expensive relative to other goods, consumers tend to **buy less X and switch to alternatives**.
- Example: if coffee becomes more expensive but tea does not, some consumers switch from coffee to tea.

2. Income effect 

- A price rise reduces the consumer's **real purchasing power**.
- Income has not changed, but the same amount of money can now buy less.
- Example: you still have £100, but higher prices mean that £100 buys fewer goods.

- Substitution effect = change caused by relative prices
- Income effect = change caused by purchasing power
- **Normal good:** both effects reduce consumption when price rises.
- **Inferior good:** substitution effect reduces consumption, but income effect may increase it.

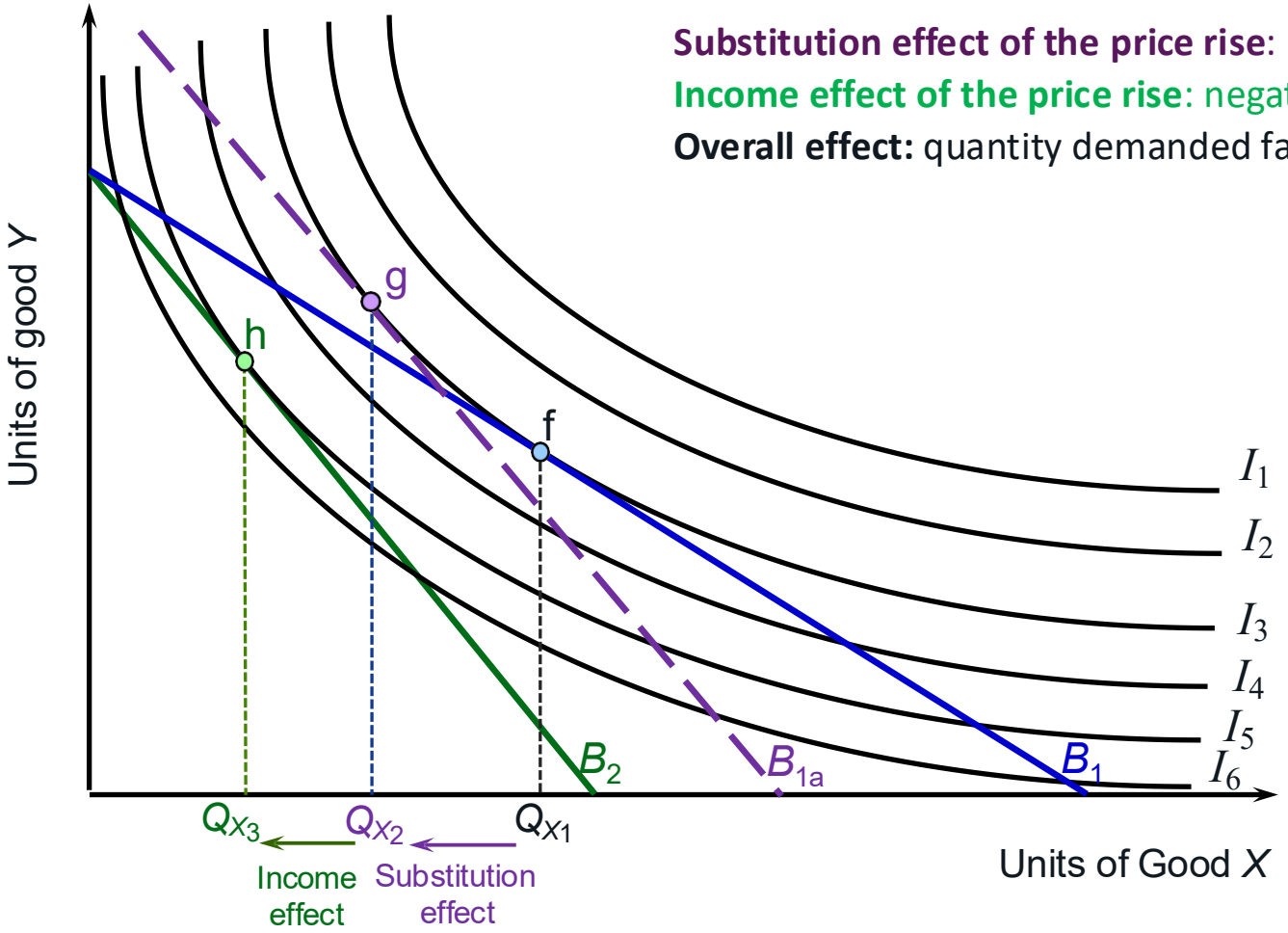
Income and Substitution Effects: Normal Good



The total effect of the price rise: consumption of X has fallen (from Qx1 to Qx3).

But this total change contains both a substitution effect and an income effect.

Normal Good: Separating the Two Effects



Substitution effect of the price rise: negative.

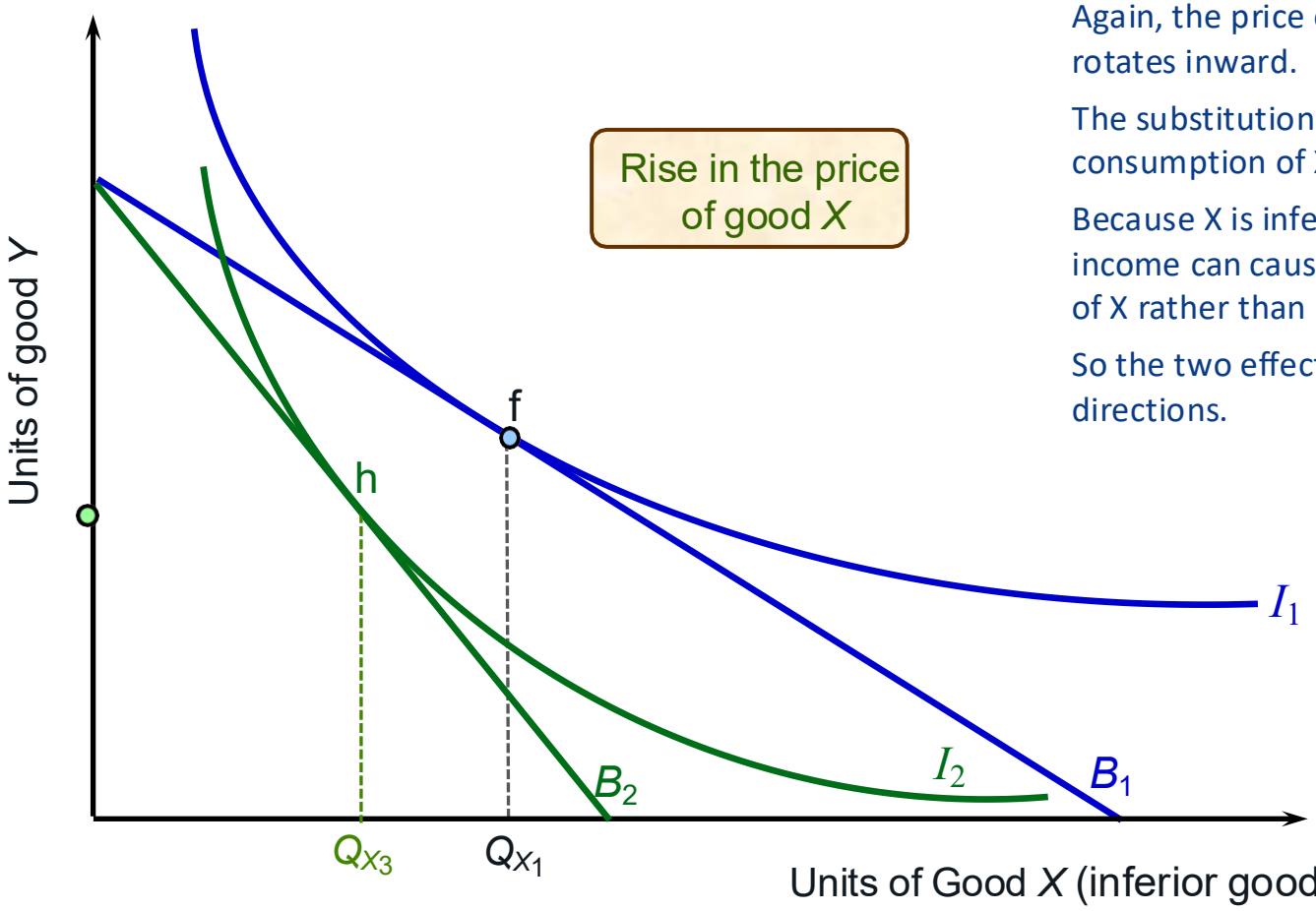
Income effect of the price rise: negative for a normal good.

Overall effect: quantity demanded falls.

First, move from f to g. This isolates the substitution effect. X has become relatively more expensive, so the consumer substitutes away from X.

Then move from g to h. This is the income effect. The price rise has reduced real purchasing power.

Income and Substitution Effects: Inferior Good



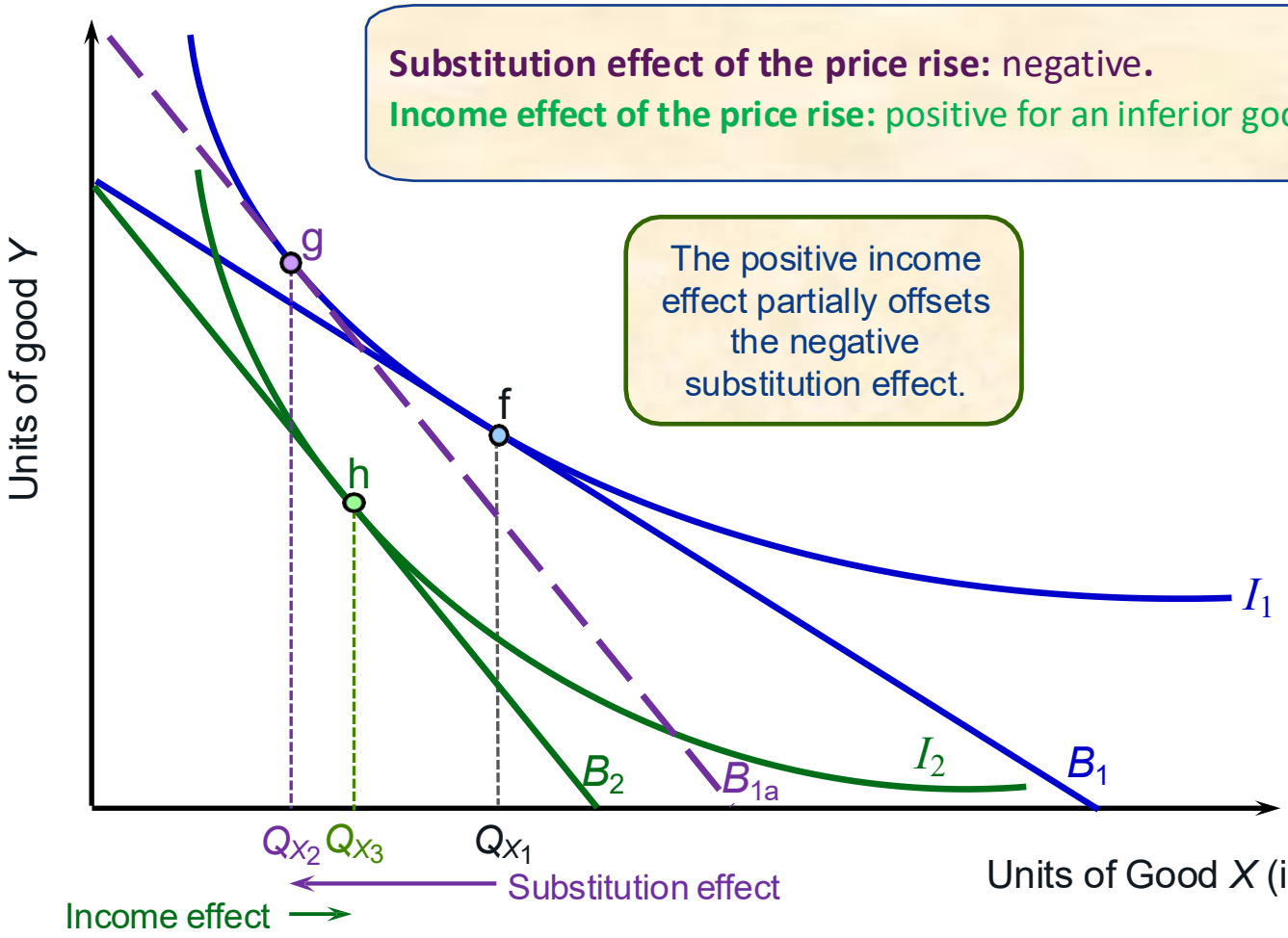
Again, the price of X rises and the budget line rotates inward.

The substitution effect is still going to reduce consumption of X. That part does not change.

Because X is inferior, a reduction in real income can cause the consumer to want more of X rather than less.

So the two effects now work in opposite directions.

Inferior Good: Separating the Two Effects



The substitution effect is negative: because X has become relatively more expensive, the consumer tends to consume less X.

But the income effect is positive: the price rise makes the consumer effectively poorer, and because X is an inferior good, that encourages more consumption of X.

Key Takeaways

What to remember from Lecture 2

Elasticity

Elasticity measures responsiveness to changes in market conditions.

Consumer choice

Demand can be linked to utility, preferences and budget constraints.

Indifference analysis

Consumers choose between combinations of goods, given preferences and budget limits.

Income and substitution effects

A price change affects consumption through both substitution and income effects.

Next step: apply these concepts in seminars using diagrams and numerical examples.

References

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- Gregory, Mankiw, N. (2017) “Economics” Cengage Textbooks
- Symeonidis, Lazaros (2022) IC103 Introductory Economics for Business and Finance materials